A mergeable measure of sentiment

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Sentiment scores for text are usually reported as one bounded number per document, such as a class probability or a normalised sum of word values. Such numbers cannot be recombined: the score of a chapter does not follow from the scores of its sentences, so long texts are cut into chunks and averaged, and the result depends on where the cuts fall. Here a measure of sentiment is introduced, the felicific signature, whose value for a span of text is computed exactly from the values of its parts when the text is divided between sentences. Each word carries positive charge (hedons) and negative charge (dolors) taken from human word ratings, and a span is summarised by 19 numbers: counts, position-weighted sums of charge, and the charge under negation and other operators. One pass over a book therefore yields the signature of every sentence, every chapter and the whole. Among position weights that admit such a merge, polynomials are the only continuous functions of relative position. On five corpora with continuous human ratings the calibrated score correlates with the ratings at $r = 0.51$, comparable to VADER (0.49), the strongest of the three other lexicon methods tested, and below two off-the-shelf transformer classifiers (0.59 and 0.68), at 4,500 and 8,700 times their speed on one processor core. On whole film reviews a simpler lexicon method agrees better with the critics' ratings, and wherever agreement above the sentence could be checked, averaged sentence scores from another method tracked the human ratings better than the merged signature. In 60 novels a conventional chunked score moves from 0.11 to 0.76 as the chunks grow, whereas merged signatures do not change. The hypothesis that sentiment is front-loaded is not supported. Words carrying no sentiment dilute a rating: the planned test supports this in four of six corpora, and in exploratory analyses of human phrase ratings the effect appears from either side and a part's weight grows with its share of the words. Compared with student essays on the same assignments, essays written by four language models close on a more positive note, as predicted, and, in an exploratory comparison, place less of their charge after intensifiers. The signature gives the same result however a text is divided between sentences, at a small fraction of a transformer's cost; on single texts, transformer classifiers remain more accurate.

Sentiment analysis turns a piece of writing into a number that says how positive or negative it is. Such numbers are computed for product and film reviews [1], for posts on social media, often millions at a time [2,3], and for parliamentary speech [4], and dozens of methods exist for computing them [5]. Two families of method are in use. Lexicon methods look each word up in a list of human word ratings and combine the values by rule [6–9]. Classifiers, now mostly fine-tuned transformers, are trained on labelled text and return a class probability [10–12].

Both families return a bounded number for each text: a probability, a mean, or a sum squashed into the interval from −1 to 1. A bounded score is intensive in the sense that temperature is intensive. It does not grow with the amount of text, and the value for a whole cannot be computed from the values of its parts unless the counts behind them are kept; a class probability and a squashed sum have none to keep. Three practical problems follow. A transformer of the BERT family reads at most 512 tokens at a time [10], so a long document is cut into chunks whose scores are averaged, and the average depends on where the cuts fall. A squashed sum saturates: in 60 of the novels studied below, VADER's compound score is beyond ±0.99 for 87% of whole chapters. And an average of averages can rank two bodies of text in the opposite order to their pooled values, the reversal known from Simpson's paradox [13–15]. The third problem is a risk that the data below barely realise: a novel's pooled density and the mean of its chapter densities order 3.1% of the 63,190 pairs of novels differently, and none of the six comparisons between film critics reverses when their ratings are held fixed.

Here sentiment is treated as a quantity that adds. Every word carries a positive and a negative charge taken from human word ratings. A span of text is summarised by a short vector, its felicific signature, named after the felicific calculus, the name later given to Bentham's proposal to measure and add pleasures and pains [16]. The signature holds counts and sums of charge weighted by powers of word position. Signatures of adjacent spans merge exactly into the signature of the joined span, so a novel is scored in one pass and every chapter, paragraph and sentence can be read off afterwards without scoring anything again. Scores that people can compare, such as net charge per thousand words or a predicted human rating with an error bar, are read from the signature at whatever scale is wanted.

The parts have precedents. The hedonometer averages human word ratings [2,6]. Word-count tools and studies of mood keep positive and negative words apart [3,9,17], in line with evidence that the two are separable in evaluation [18]. Political scientists score texts from counts of opposed items on a log-ratio scale with a stated uncertainty [4,19]. Summaries that merge exactly across partitions, including sums weighted by polynomial or exponential functions of time, are standard in stream processing [20–23]. A model that composes the sentiment of a phrase by multiplying matrices assigned to its words is associative in the same way [24]. Emotional arcs of stories have been drawn with moving windows and transforms [25,26], and features of such arcs have been used to predict judgements of literary quality [27]. This paper joins these into one object for sentiment, states which position weights such an object can carry, and tests it against human ratings.

The aim is a measure for which granularity holds: the score of a whole is the merge of the scores of its parts, at every level from the sentence to the book (Fig. 1). Two empirical questions about how sentiment is distributed in text are tested along the way, each stated with a decision rule before any evaluation data were scored (Methods). Front-loading holds that words carrying sentiment come early in a text and weigh more there; dilution holds that words carrying none weaken the judged sentiment of the whole. Front-loading is not supported: charged words gather at the ends of sentences, tweets and posts and are spread almost evenly through reviews and chapters, and whether early words weigh more with readers depends on the genre. An alternative specified in advance, recency, met its planned tests, but two of them

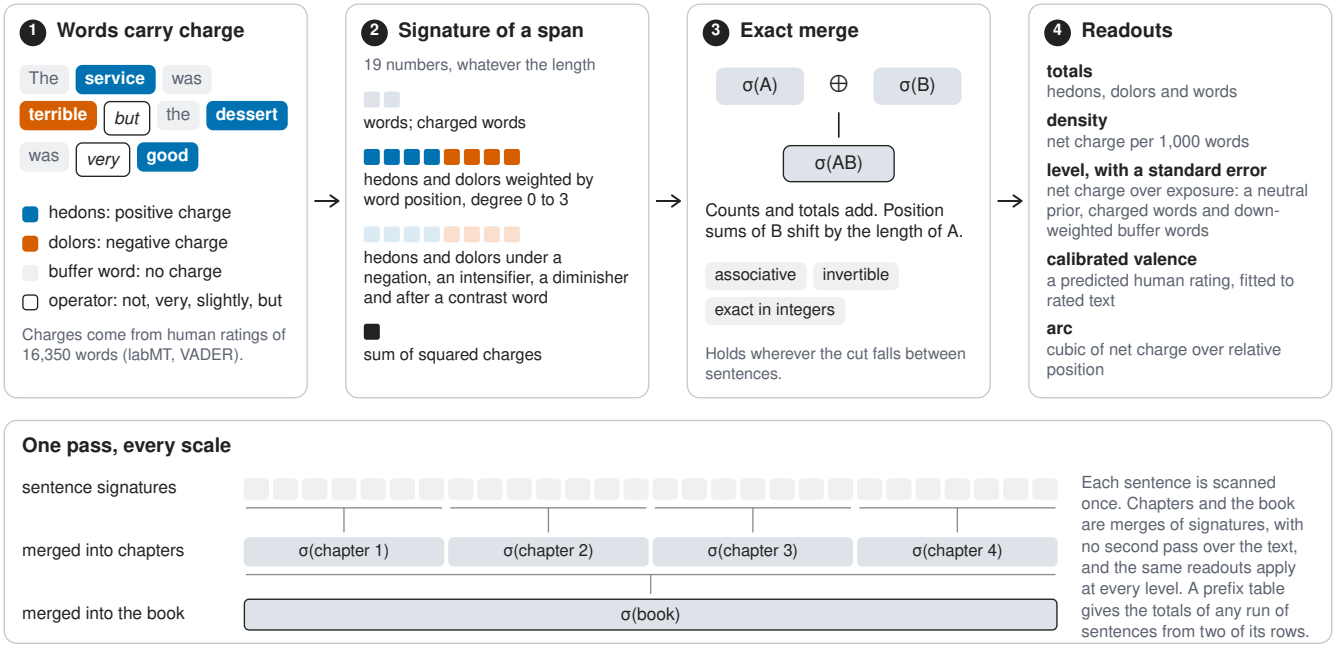


Figure 1 The felicific signature. 1, Each word carries positive charge in hedons (blue) or negative charge in dolors (vermillion), taken from human word ratings; words rated near neutral are buffer words with no charge (grey), and operator words (outlined) carry no charge themselves; the charge of each word in their reach is counted as usual and also recorded in that operator's entries of the signature, which only the calibrated valence uses. The colours of the example are those of the released lexicon. **2**, The signature of a span: two counts, sums of hedons and dolors weighted by powers of word position, the charge under each of four operators, and the sum of squared charges. **3**, Signatures of adjacent spans merge exactly when the first ends at a sentence end: counts and totals add, and the position sums of the second span shift by the length of the first. **4**, Scores read from a signature. Bottom, a text is scanned once, sentence by sentence; the signatures of chapters and of the whole are merges, with no second pass over the text.

confound order with length and reverse once length is held fixed. Dilution is supported by the planned test in four of six corpora, and exploratory analyses of human phrase ratings show it on either side of the charged words. Finally, the measure is applied to essays written by language models and by students on the same assignments.

Results

A signature that merges

Take a text as a sequence of words. Word i has a positive charge h_i and a negative charge d_i, each between 0 and 1, at most one of them nonzero. The values come from two open sets of human word ratings, the labMT list[2] (10,222 words, 50 raters each) and the VADER list[7] (7,494 entries, 10 raters each). On the 1,366 words they share, the two sets correlate at $r = 0.95$. Both are put on one scale, on which 1 is the positive end of the rating scale and 0 its neutral point, and are averaged where both rate a word (Methods). Positive charge is measured in hedons (hd) and negative charge in dolors (dl), the names that moral philosophy uses for units of pleasure and pain[28,29]. One hedon is the charge of a word that every rater places at the top of the scale; *love* carries 0.85 hd and *terrible* 0.55 dl. A word rated within a quarter of the scale from neutral gets no charge and is called a buffer word, following the neutral band of the hedonometer[2]. In the six rated corpora used here, 23% of running words are charged.

The signature of a span holds the number of words L, the number of charged words E, and the sums

$$S_k^+ = \sum_i h_i\, i^k, \qquad S_k^- = \sum_i d_i\, i^k, \qquad k = 0, \ldots, 3, \tag{1}$$

where i is the position of the word in the span, counted from zero. S_0^+ and S_0^- are the span's total hedons and dolors. The signature also holds the hedons and dolors that fall under a negation, after an intensifier, after a diminisher, and after a contrast word such as *but*, together with the sum of squared charges (Methods). That is 19 numbers, whatever the length of the span (Fig. 2a,b).

Signatures merge. If span A ends at a sentence end and is followed by span B, the counts and totals add, and the position sums of B are shifted by the length of A:

$$S_k(AB) = S_k(A) + \sum_{j=0}^{k} \binom{k}{j} L_A^{k-j}\, S_j(B). \tag{2}$$

The operation is associative and can be undone, so the signature of B follows from those of AB and A (Proposition 1, Methods). At a cut inside a sentence the counts, the totals and the position sums still merge exactly, and the operator entries can differ, because the reach of an operator may cross the cut. A text of any length is therefore scored in one pass with constant memory. Parts can be scored separately and merged in reading order to the same result as a single pass. A table of prefix signatures, one row per sentence, gives the signature of any contiguous run of sentences from two of its rows. In integer arithmetic these statements hold to the last digit, and the reference implementation, which counts in thousandths of a hedon, is tested for exact equality on random texts. The fast implementation uses double-precision floats; over 356 novels its merged and single-pass signatures agree to a relative error below 2×10^{-13}. In floats the two-row query is accurate at any depth for counts and totals, and so for density and level. The position sums of a short run deep in a long text are lost to cancellation, so the arc or the predicted rating of such a run is read by merging the run's own rows, at a cost that grows with its length.

Not every position weight allows a merge. Suppose a summary keeps finitely many sums $\sum_i c_i\, \varphi(i)$ of charge against weight functions φ. For charges that may take any real value, the summary of AB

can be computed from the summaries of A and B and their lengths exactly when the weight functions span a space that is closed under shifting the position. Such weights are combinations of $i^k \rho^i$, with ρ real or complex (polynomials, exponentials, oscillations and their products), and of weights that vanish after the first few positions (Proposition 2, Methods). The statement is an instance of a classical fact about translation-invariant function spaces[30,31] and about functions on strings with finite-dimensional representations[32,33], and stream-processing systems rely on it[20]. A decay factor, a weight that falls geometrically with position, is therefore one of the few position weights that survive a merge, at the cost of one number per decay rate. And among mergeable weights, polynomials are the only ones that give a continuous function of relative position, that is, of position divided by a length that is unknown until the text ends (Methods). The signature keeps polynomial sums for that reason: the sum of charge under any cubic function of relative position can be read from it, for a post and for a novel alike.

Reading a score from a signature

A signature is extensive. Scores that can be compared between texts are read from it (Fig. 2c).

Totals are the hedons $H = S_0^+$ and dolors $D = S_0^-$ over L words. *Density* is the net charge per thousand words, $1000\,(H - D)/L$; the hedonometer's average happiness is a density taken over charged words only.

Level follows a model from the psychology of judgement. In Anderson's averaging model a judgement is a weighted mean of the pieces of information received and an initial neutral impression[34], and information that says nothing pulls a judgement toward neutral, the dilution effect[35]. For words this gives

$$\text{level} = 1000\,\frac{H - D}{a + E + \kappa B}, \tag{3}$$

where $B = L - E$ counts buffer words, a is the weight of the neutral first impression in units of charged words, and κ is the weight of a buffer word relative to a charged one. With $\kappa = 0$ buffer words do nothing; with $a = 0$ and $\kappa = 1$ the level is the density. The denominator is the exposure, the amount of evidence behind the score, and it gives the level a standard error that shrinks as a text grows (Methods).

Calibrated valence is a predicted human rating. It is a linear combination of ten signature entries, each divided by the exposure: hedons, dolors, three position moments of net charge, negated hedons, negated dolors, and the net charge that is intensified, diminished, and after a contrast word. Given a and κ, the coefficients solve a ridge regression against human ratings, a convex problem with a closed-form solution; a and κ are chosen on a grid (Methods). The fitted coefficients say how much a negation flips a word and how position is weighed; they are estimates with intervals, in the tradition of rule constants measured from ratings[7,36].

The *arc* is a cubic fit of net charge against relative position. It has a closed form in the position sums and equals the least-squares cubic up to terms that shrink as the inverse square of the span's length (Methods), so it costs nothing beyond the signature and suits spans of a few dozen words or more.

Agreement with human ratings

Scores were compared with human ratings on six corpora: sentences from six genres (EmoBank[37]), tweets (SemEval-2018[38]), Facebook posts[39], film-review sentences (the Stanford Sentiment Treebank, SST[40]), whole film reviews with the critic's rating[41], and Yelp reviews with star ratings[42]. The first five carry continuous ratings and form the primary set. Every constant was fitted leave-one-corpus-out: for each target corpus, a, κ and the ten coefficients came from the other five, and scores were then computed on the target's held-out test portion (Methods). The six fits chose a of 2 or 4 and κ of 0.2 or 0.5.

Averaged over the five primary corpora, the calibrated score correlates with human ratings at $r = 0.51$, against 0.49 for VADER[7], 0.44 for TextBlob[43] and 0.42 for the hedonometer[2] (Fig. 3a; all values with intervals in Extended Data Table 1). The plan set one criterion for accuracy: a mean r at least that of VADER (Methods). It is met, 0.506 against 0.489, by a margin that is within noise. The mean difference is +0.016 with a t interval over the five corpora of [−0.017, +0.049], and on the plan's secondary outcome, Spearman's ρ, the two are level (0.506 and 0.508; Extended Data Table 2). Corpus by corpus, the paired difference from VADER excludes zero in favour of the signature in three of the six corpora and in favour of VADER in none. Two of the three are the corpora of long reviews, where VADER's score for a whole text saturates. Scored sentence by sentence and averaged, as its documentation recommends for longer text (a comparison added after review), VADER rises to 0.30 on film reviews and 0.63 on Yelp. Against this version the signature leads in one corpus (Facebook posts) and trails in one (Yelp, −0.017 [−0.032, −0.002]), and the five-corpus means are 0.51 and 0.49. On whole film reviews, the longest rated texts, TextBlob does better than either ($r = 0.46$ against 0.29 for the signature; difference −0.168 [−0.200, −0.136]). The level of equation (3), whose only fitted quantities are its own a and κ (a of 1 or 2 and κ of 0.2 across the six fits), reaches 0.48, below VADER.

Methods given more labelled text do better. A bag-of-words ridge regression trained on the same five-corpus pool as the signature's constants, a comparator added after the first results, averages 0.55, and trained on each corpus's own training portion 0.61. Two transformer classifiers, used as distributed, average 0.59 (DistilBERT fine-tuned on SST-2[12]) and 0.68 (a RoBERTa model fine-tuned on tweets[44]). These figures understate what the models can do: long reviews were cut at 512 tokens, and tweets were not given the handle masking that the RoBERTa model's documentation prescribes (Methods). Even so, their lead varies by corpus: the signature is ahead of DistilBERT on tweets and on Facebook posts (differences +0.103 [+0.062, +0.145] and +0.077 [+0.036, +0.115]), level with it on EmoBank, and far behind on SST, the domain DistilBERT was tuned on, and on reviews. The signature is not a replacement for these models where accuracy on one text matters most.

Removing parts of the signature shows what earns its place (Fig. 3c; means over all six corpora). Without the operator channels the mean correlation falls by 0.034, and without dilution ($\kappa = 0$) by 0.015. Without the neutral prior ($a = 0$) the mean falls by 0.010: the prior helps in two corpora, costs a little in film reviews and makes no clear difference in the other three. Merging hedons and dolors into one channel costs 0.002 and dropping the position moments 0.000: neither matters for agreement with raters, though both are needed for the readouts below.

The fitted coefficients (Extended Data Table 4) can be read as rules, in an analysis the plan did not list and with a caution: the readings belong to one pooled fit, the shipped model, and the bootstrap intervals describe sampling error within that fit and say nothing about other corpora. In it a dolor moves the predicted rating 1.63 [1.54, 1.73] times as far as a hedon. A negated positive word counts as −1.60 [−1.85, −1.37] times its charge and a negated negative word as −0.01 [−0.23, +0.20]: *not good* is bad, and *not bad* is neutral. An intensifier adds +1.02 [+0.85, +1.19] times the charge of the word it modifies. Across the six leave-one-corpus-out fits the first three readings run from 1.17 to 1.81, from −1.95 to −1.10 and from −0.14 to +0.10, and fits to single corpora differ far more (Extended Data Table 5): the ratio

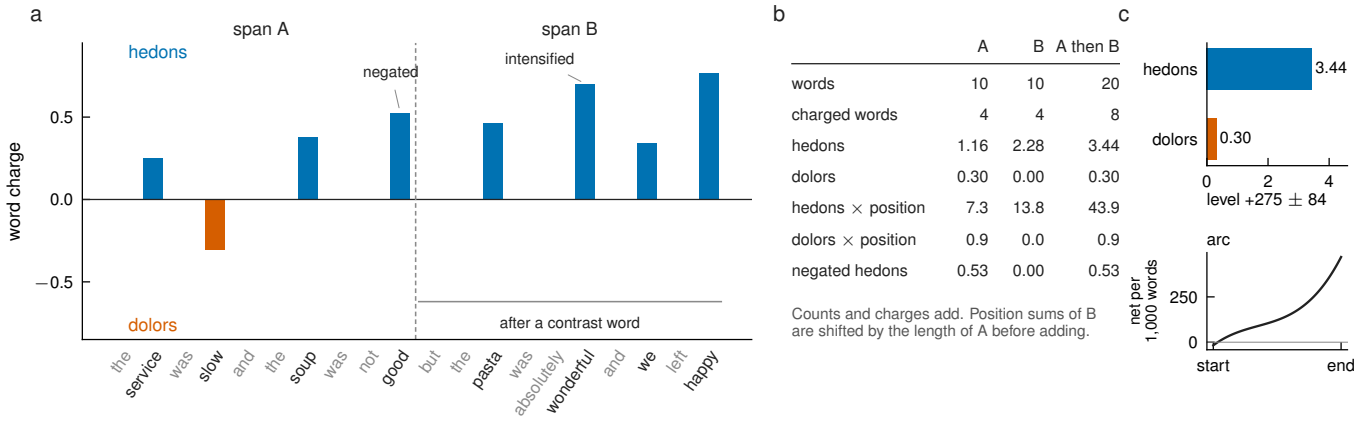


Figure 2 Charge, signature, merge, readouts. a, Two sentences scored word by word. Bars above the line are hedons, bars below are dolors; grey words are buffer words and carry no charge. *Good* sits inside the reach of *not*, *wonderful* follows an intensifier, and the second sentence follows a contrast word. **b**, Seven of the nineteen entries of the signature of each sentence and of the two read as one text, obtained by merging and equal to a single pass over both. **c**, Readouts of the merged signature: totals, the level with its standard error (the level's own constants, $a = 1$ and $\kappa = 0.2$), and the cubic arc of net charge over relative position. Totals and level ignore the operator entries, so *good* under *not* still counts as positive here; only the calibrated valence uses them.

of dolor to hedon is below 1 in SST and above 4 in Yelp. Two readings have the same sign in every fit, pooled or single: a negated positive word turns negative, and an intensifier adds to the charge of the word it modifies. The sizes belong to these corpora. The data do not settle the effect of diminishers (−0.33 [−0.91, +0.09]) or of contrast words (−0.06 [−0.14, +0.02]); in particular they give no support to the rule that the clause after *but* counts for more[7].

Word charges can also be fitted, in an analysis the plan did not list. With the exposure and the operator constants held fixed, the predicted rating is linear in the vector of charges, so the charges that best predict ratings, pulled toward the human word norms by a ridge penalty, solve one convex problem (Methods). Fitted leave-one-corpus-out, these refined charges raise the mean correlation to 0.55, the same as the bag-of-words ridge trained on the same five corpora (0.55). The gain is uneven. It is large on review text (SST 0.44 to 0.55, film reviews 0.29 to 0.49) and negative on the mixed genres of EmoBank (0.48 to 0.41) and on Facebook posts (0.62 to 0.58). Charges learned from ratings carry the genre of the texts they were learned from, and their attitudes. Of 20 words checked, words for groups of people and the names of two public figures, 14 moved down by more than 0.05 on the charge scale (*women* from +0.53 to −0.08, *muslim* from −0.09 to −0.37; Extended Data Table 7). Human word norms travel better, and the refined charges are not part of the released package.

A check that the plan did not list shows that the standard error of a level does not track the error against human ratings. It shrinks with exposure, and the error of the calibrated score does not follow it. From the lowest to the highest quartile of exposure the error falls in two corpora (EmoBank and Yelp) and rises in the other four (Extended Data Fig. 1a). The standard error says how well a text's own words pin down its level. It is not a prediction interval for a human rating.

Cost

Scoring is one table lookup per word. On one core of a rented six-core desktop processor the implementation, written in Python with NumPy, converts 167,000 sentences (2.6 million words) per second into signatures and levels (Fig. 3b). On the same core VADER scores 27,900 sentences per second, TextBlob 11,100, DistilBERT 37 and the RoBERTa model 19. The signature is therefore about 4,500 times faster than DistilBERT and 8,700 times faster than the RoBERTa model, well past the factor of 100 that the plan set as its criterion. On a graphics card (RTX 3090, half precision) DistilBERT reaches 12,200 sentences per second, and one processor core running the signature is still 14 times faster. Time grows in proportion to the amount of text (log-log slope 1.02 from 10^4 to 10^6 sentences); ten million sentences took 59 seconds in fixed-size batches. Because signatures merge, the work divides among processes without changing any result, and twelve processes on the six cores scored 0.8 million sentences per second, 4.8 times the rate of one. At these rates a billion sentences take 1.7 hours on one core or 21 minutes on the six, against 23 hours for DistilBERT on the graphics card and 310 days on one core.

Where the feeling sits

The plan drew three predictions from front-loading: that charged words come early, that readers weigh early words more, and that rated intensity falls from the first sentence of a document to the last (Fig. 4).

Charged words do not come early. The charge-weighted mean position, on a scale from −1 at the first word to +1 at the last, is above zero in all eight kinds of unit examined (the six rated corpora, paragraphs of the film reviews and chapters of 150 of the novels), with every interval excluding zero (Extended Data Table 3). The size differs by kind. In EmoBank sentences, SST sentences and tweets the first tenth of the words carries 0.65 to 0.75 of the text's mean charge per word, and in those three and in Facebook posts the last tenth carries 1.34 to 1.55 (Fig. 4a): English sentences tend to open with determiners and pronouns and close on content. In reviews, paragraphs and chapters the charge is spread almost evenly, with mean positions from +0.002 in Yelp reviews to no more than +0.03. Yelp reviews are the one place with a charged opening, at 1.09 times the mean, matched by the close at 1.11. Negative charge sits later than positive charge in seven of the eight.

Whether readers weigh early words more depends on the kind of text. The position kernel fitted in each corpus puts more weight on the first third of a text than on the last third in SST sentences (difference +0.26 [+0.14, +0.39]), as front-loading predicts. It puts less in tweets (−0.39 [−0.54, −0.24]) and in Yelp reviews (−0.12 [−0.15, −0.09]), and is undecided in the other three (Fig. 4c). In the shipped model, fitted on all six corpora, the last third outweighs the first (−0.13 [−0.20, −0.07]).

The third prediction is undecided. Within EmoBank documents, rated intensity neither falls nor rises with sentence position (slope +0.012 [−0.012, +0.032] per document length; 115 documents).

Front-loading is therefore refuted as a claim about where writers put feeling words. As a claim about what readers weigh it holds in film-review sentences, fails in tweets and in Yelp reviews, and is

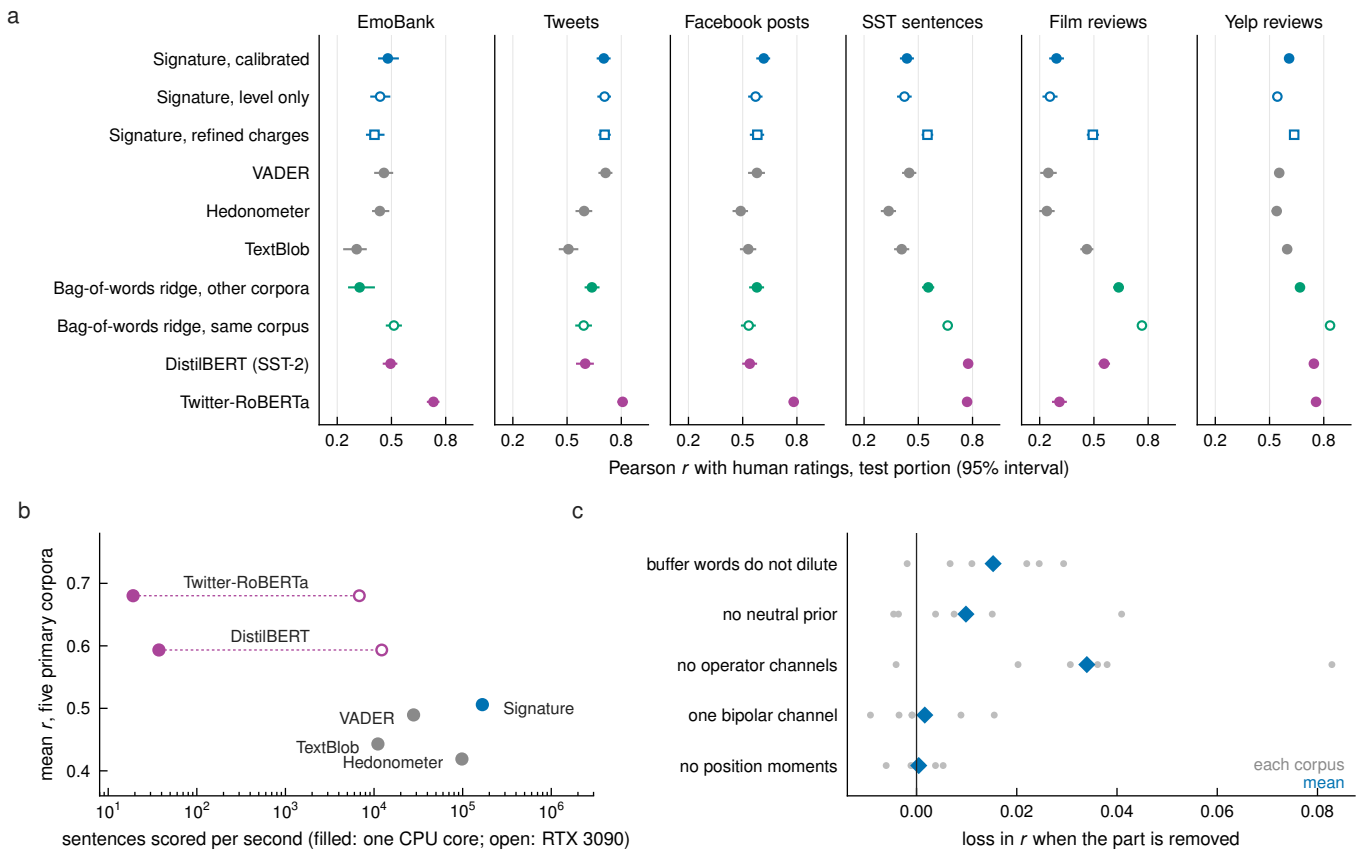


Figure 3 Agreement with human ratings, and cost. a, Pearson r between score and human rating on the held-out test portion of each corpus, with 95% bootstrap intervals. Blue: readouts of the signature, with constants fitted on the other five corpora (filled circle, calibrated valence; open circle, level; square, calibrated valence with refined word charges). Grey: lexicon methods. Green: bag-of-words ridge regression trained on the other five corpora (filled) or on the same corpus (open). Purple: transformer classifiers. **b**, Mean r over the five corpora with continuous human ratings against sentences scored per second on one CPU core (filled) and, for the transformers, on a graphics card (open). **c**, Loss in r when one part of the signature is removed and the constants are refitted.

undecided elsewhere.

Order and length in human phrase ratings

The SST gives a human rating to every phrase of its parse trees[40], so the rating of a phrase can be regressed on the ratings of its two parts. In the training trees the later part gets the larger weight, 0.68 against 0.61 where both parts have at least two words (29,636 phrase pairs). Together with charged words gathering at the ends of sentences, this suggests a hypothesis opposite to front-loading: recency, that what comes last in a text counts most toward a reader's judgement of it. Three tests of it were specified before the data they concern were examined, the development and test trees of the treebank and 22,001 IMDB film reviews[45] (Methods). All three met their stated criteria. In 11,474 held-out phrase pairs the later part has weight 0.69 and the earlier part 0.63 (R1; difference +0.059 [+0.032, +0.085]). A rated phrase keeps 0.68 of its rating when two or more neutral words follow it and 0.77 when they come before it (R2; difference +0.091 [+0.055, +0.127]). And the last fifth of an IMDB review separates favourable from unfavourable reviews better than the first fifth does (R3; area under the curve 0.696 against 0.657; difference +0.039 [+0.031, +0.048]).

R1 and R2, however, do not hold length fixed. The later part of a phrase is the longer one in 62% of pairs (7.8 words on average against 4.7; all trees, 41,110 pairs), and the weight of a part rises with its share of the phrase's words, from 0.29 for a part with under a tenth of them to 0.86 for a part with over nine tenths (Fig. 4b). Where the two parts are equally long, the earlier one has the larger weight: 0.71 against 0.57, a difference of −0.138 [−0.182, −0.090] for the later part (5,366 pairs), and −0.164 [−0.257, −0.081] in the held-out trees that R1 used. With one curve of weight against word share for both parts, the later part's extra weight is −0.019 [−0.035, −0.002] (held out, −0.030 [−0.059, −0.002]). R2 fails the same way. The rated parts that were followed by neutral words averaged 3.8 words and those preceded by them 9.1, and a short rated part keeps less of its rating whichever side the neutral words are on. Matched on both lengths in the held-out trees, with two or more neutral words as in R2, the share kept is lower when the neutral words come first (before minus after, −0.075 [−0.118, −0.032]; −0.050 [−0.064, −0.035] when one-word neutral parts are included).

So in film-review phrases the longer part counts for more and, length for length, the earlier part counts for more: by 0.14 when the two parts are equally long, and by 0.02 under one curve of weight against word share. This agrees with the kernel for SST sentences above. English syntax may be the cause as much as order, since the two parts of a phrase differ in grammatical role. Weight rises with share, yet weighting by length does not improve the fit: a plain mean of the two part ratings accounts for the phrase rating as well as a length-weighted mean (R^2 = 0.59 and 0.58; 0.59 with free weights).

R3 compares fifths of equal length, so length does not explain its result, and it measures where reviewers place their verdict, which is a fact about writers. The scorer's density in the first, the three middle and the last fifth separates the two classes with areas of 0.657, 0.648

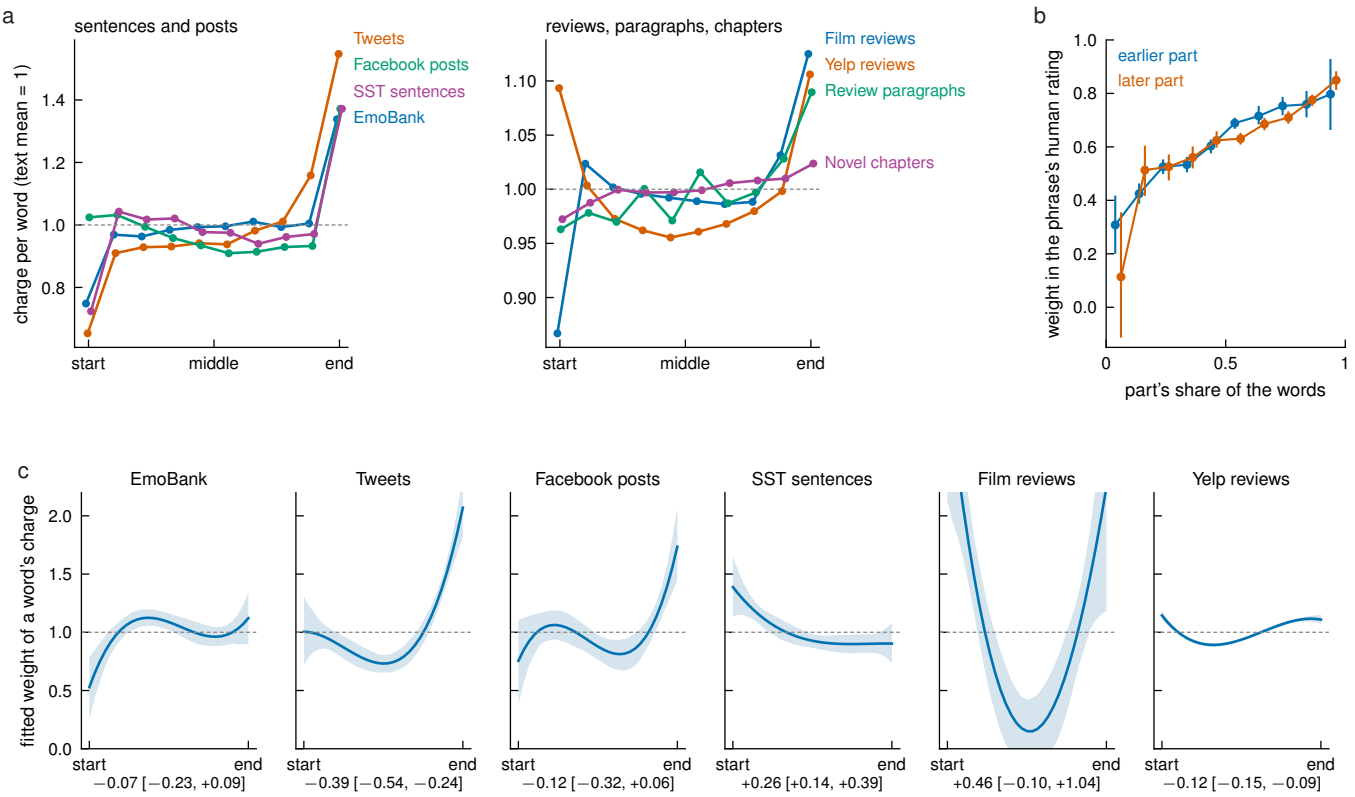


Figure 4 Where feeling sits, and what readers weigh. a, Charge per word in each tenth of a text, relative to the text's mean, for texts of ten or more words. **b**, Weight of a part in a regression of a phrase's human rating on the human ratings of its two parts, by the part's share of the phrase's words, for earlier and later parts of 41,110 binary nodes of the Stanford Sentiment Treebank (95% intervals, sentences resampled). Weight rises with share for both parts. **c**, Position kernel fitted against human ratings in each corpus: the weight given to a word's charge as a function of its relative position, with 95% bands. Below each panel, the mean weight over the first third minus the mean over the last third. The kernel for whole film reviews is poorly determined and runs off the scale.

to 0.651 and 0.696: the verdict shows at both ends of a review and most at the end, as earlier work on reviews found[46–49].

Recency is therefore not supported as a claim about readers: R1 and R2, the two tests that concern readers, were met only because they compared parts of unequal length.

Neutral words dilute, from either side

The planned test of dilution (Fig. 5) asks whether a readout with a buffer weight $\kappa > 0$ agrees with held-out ratings better than the same readout with $\kappa = 0$. It does in four of six corpora (tweets, Facebook posts, SST and Yelp) and is undecided in EmoBank (+0.007 [−0.003, +0.020]) and in film reviews (−0.002 [−0.009, +0.004]). Two statistics that the plan did not list agree in direction. Holding a text's total charge fixed, rated intensity falls as the number of buffer words rises in five of six corpora, EmoBank among them, with partial rank correlations from −0.27 in EmoBank to −0.08 in SST sentences (Fig. 5b); whole film reviews show no effect (+0.01 [−0.02, +0.04]). In fits to single corpora, agreement is highest at a buffer weight between 0.2 and 1 in the same five and at zero for film reviews, with gains in r of 0.009 to 0.020 that carry no intervals (Fig. 5c).

In an exploratory analysis, human phrase ratings show the effect without any scorer. Take one rated word of the treebank joined to a neutral phrase of m words. With the neutral words after it, the joined phrase keeps 0.61 [0.60, 0.63] of the word's rating for one neutral word and 0.42 [0.35, 0.47] for five to eight (0.39 [0.27, 0.49] for nine or more). With the neutral words before it, the share is 0.54 [0.52, 0.55] for one neutral word and 0.16 [0.00, 0.30] for five to eight, the longest run with 30 or more phrases on that side (Fig. 5a). The share falls with m on both sides. Over rated parts of every length in all trees, matched on the lengths of both parts, the share kept is lower when the neutral words come first than when they follow (before minus after, −0.050 [−0.058, −0.042]). Because the rated part was selected for a strong rating, regression to the mean lowers every share. It does not produce the fall with m or the difference between sides.

Words that carry no feeling therefore weaken the judged feeling of the whole, and more of them weaken it more; in the phrase ratings they do so from either side, as the averaging account of impression formation predicts[34,35].

A book at every scale

The 356 novels from Project Gutenberg (28.1 million words, median 25 chapters) were scored sentence by sentence, and their signatures merged up to chapters and books. Fig. 6a,b shows *Moby-Dick*: 213,523 words carrying 11,790 hd and 3,539 dl, a density of +39 per thousand words, with each of the 129 sections that the chapter rule finds in it (the book has 135 chapters and an epilogue; Methods) and twelve spans of the last one read from the same pass.

Chapters are not random samples of a book's sentences. The variance of chapter densities is 5.9 times (median; interquartile range 4.3 to 7.8) what results when a novel's sentences are shuffled across its chapters, and exceeds the shuffled value at $p \leq 0.01$ in 98% of novels (Fig. 6c). The shuffle destroys all continuity between neighbouring sentences, so the ratio shows that tone is locally coherent in novels. It does not show that chapter boundaries are where tone changes, which would need a null that moves the boundaries and keeps the order.

The cubic arc read from a book's signature captures 41% (median) of the variance of an arc drawn with a moving window one tenth of the book wide; keeping moments up to degree nine would capture 63%

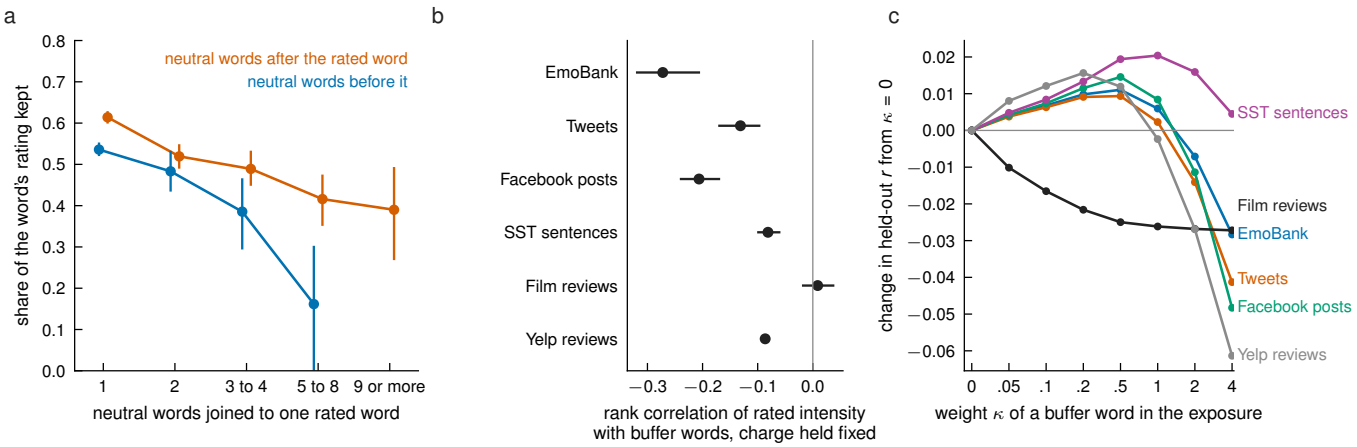


Figure 5 Neutral words dilute a rating from either side. a, Human phrase ratings from the Stanford Sentiment Treebank. One word with a clear rating is joined to a neutral phrase; the vertical axis is the share of the word's rating that the joined phrase keeps, by the number of neutral words and their side (95% intervals; cells of 30 or more phrases). **b**, Partial rank correlation between rated intensity and the number of buffer words in a text, with total charge held fixed. **c**, Change in held-out r as the weight κ of a buffer word in the exposure moves away from zero, in fits to each corpus alone with a fixed at the shipped value and coefficients refitted at each κ; the grid is extended to $\kappa = 4$.

(Fig. 6d). The two leading modes of variation among the window arcs of the 356 novels carry 22% and 19% of the variance and are close to the Legendre polynomials P_1 and P_2 (cosine similarity 0.94 and 0.89; Fig. 6e), the rise or fall and the single turn that earlier studies of story shapes found[25]; the third mode is not close to the cubic (0.10). A wide moving window yields smooth leading modes from any series, so the resemblance is weak evidence about stories.

Human labels exist for one body of narrative. Alm and colleagues labelled every sentence of 176 fairy tales with an emotion[50]. The cubic arc of a tale's merged signature correlates positively with the cubic arc of its human labels in 70% of tales (median correlation 0.46; sign test $p = 6 \times 10^{-8}$; Extended Data Fig. 1b). For a tale's overall valence the picture is less favourable. The calibrated score of the merged signature correlates with the mean human label at 0.50 [0.34, 0.63], and VADER averaged over sentences does better (0.63 [0.51, 0.73]). The human target here is itself a mean of sentence labels, which an average of sentence scores mirrors. VADER applied to the whole tale saturates beyond ±0.99 for 73% of tales and falls to 0.39.

Word norms rate words and know nothing of events. The final chapter of *Moby-Dick*, in which the ship is destroyed, scores near the book's own density, because *whale*, *sea* and *boat* are pleasant words to raters. Densities of narrative text are best read against the text's own baseline.

Pooling, averaging and chunking

A mean of chunk scores depends on how a long text is cut (Fig. 7a to c). For the first 60 novels in file order, a book score was computed as the mean VADER compound score over chunks. The mean across books rises from 0.11 with one-sentence chunks to 0.76 with whole chapters, and 19% of pairs of books swap order between the two. The same happens on 300 whole film reviews. The score of a review and the mean score of its sentences differ in sign for 28% of reviews with DistilBERT, 51% with the RoBERTa model and 5% with VADER, whose whole-review score is beyond ±0.99 for 48% of them. For the transformers the comparison is confounded by truncation: 97% of the reviews exceed the models' 512-token window, so the whole-review score reads the opening of a review and the sentence mean reads all of it. In all three cases the mean of sentence scores predicts the critic's rating better than the whole-review score does ($r = 0.61$ against 0.53, 0.52 against 0.23, and 0.23 against 0.17).

The dependence comes from averaging, whatever is averaged. If the densities of chunks are averaged, 8% of pairs of books swap order between one-sentence chunks and chunks of 125 sentences, against 11% for VADER (10% and 19% with whole chapters). If the chunks' signatures are merged, the book's density is the same for every chunking, to a relative difference below 2×10^{-14}.

A score that does not change with the cut is not for that reason accurate. On the same 300 reviews the merged density correlates with the critic's rating at 0.23, the same as VADER's sentence mean and far below the mean of DistilBERT's sentence scores (0.61), and among the fairy tales an average of VADER's sentence scores tracked overall valence better than the merged signature did.

A pooled value and a mean of parts differ by a covariance. For parts with word counts n_p and densities δ_p,

$$\frac{\sum_p n_p \delta_p}{\sum_p n_p} = \bar{\delta} + \frac{\mathrm{Cov}(n, \delta)}{\bar{n}}, \qquad (4)$$

with $\bar{\delta}$ and $\bar{n}$ the unweighted means over the P parts and $\mathrm{Cov}(n, \delta) = P^{-1} \sum_p (n_p - \bar{n})(\delta_p - \bar{\delta})$, an identity familiar from the decomposition of aggregate productivity[51]. The two differ when long parts differ in density from short ones. In novels they rarely do: the median gap between a novel's pooled density and the mean of its chapter densities is 4% of the standard deviation of density between novels, and 3.1% of the 63,190 pairs of novels are ordered differently by the two (Fig. 7d).

A difference between two bodies of text splits in the same spirit. Kitagawa's decomposition[52] writes a difference in pooled density as a part due to differences within strata plus a part due to the mix of strata. Four film critics in the review corpus[41] illustrate it, with the critic's own rating as the stratum (Fig. 7e). Pooled, Berardinelli's and Renshaw's reviews have the same density (difference −0.1 [−1.3, +1.1] per thousand words). Within the same rating the difference is −1.2 [−2.5, +0.0], an interval that reaches zero, and the mix of ratings contributes +1.1 [+0.8, +1.6]. None of the six pairs of critics reverses between the pooled and the within-rating comparison. The example shows the bookkeeping and supports no claim about the critics' prose: the reviews were scored as distributed, with each critic's credits and sign-off lines left in, and one critic's standard footer alone accounts for about half of a difference of this size (Methods). Which part of a decomposition answers a question depends on the question[53]; a signature kept per stratum makes both parts available, and a single pooled score of any kind does not.

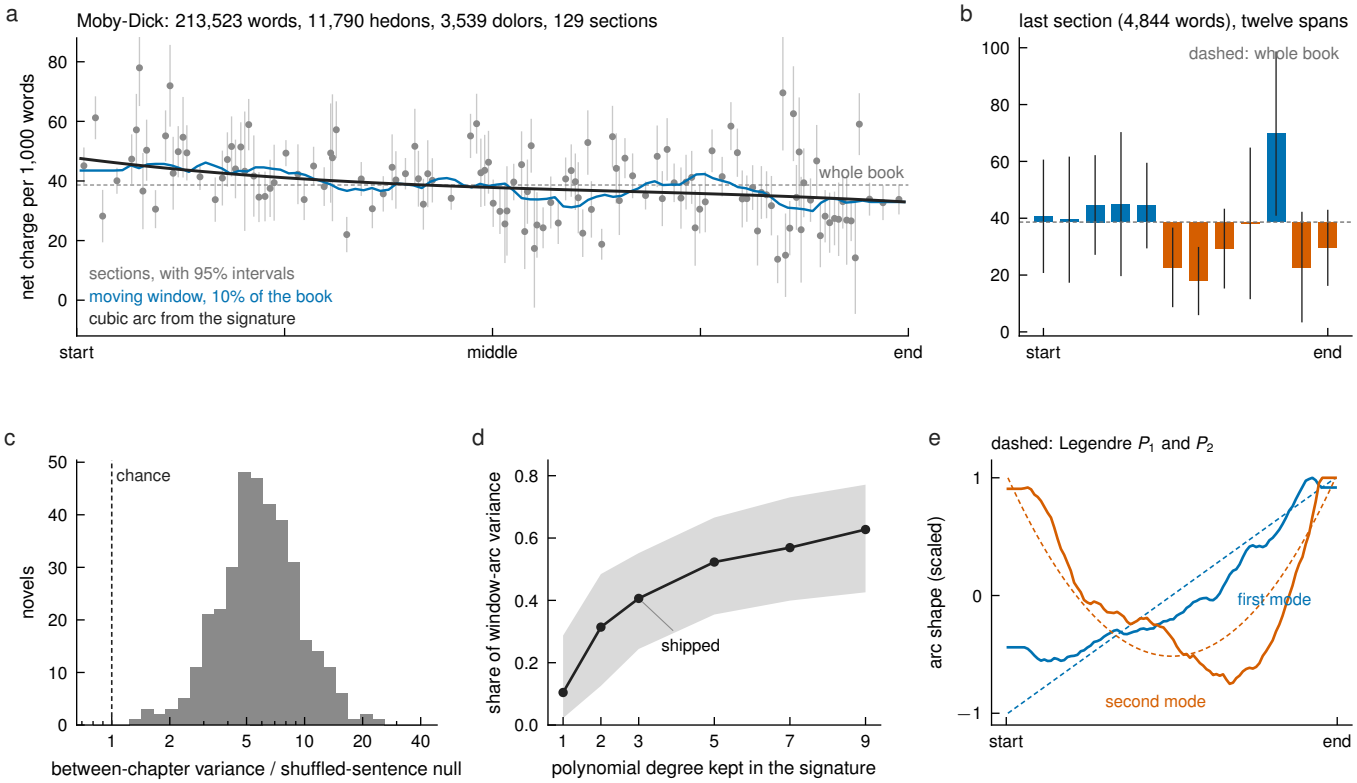


Figure 6 A novel at three scales from one pass. a, *Moby-Dick*: density of each of the 129 sections found by the chapter rule, with 95% intervals from the word-level standard error (too narrow where charge clusters), the cubic arc read from the book's signature, and a moving-window arc for comparison. **b**, The last section (the final chapter and the epilogue) cut into twelve runs of sentences, each read from two rows of the section's prefix table; bars show the difference from the whole book's density, with intervals as in **a**. **c**, Between-chapter variance of density relative to its value when sentences are shuffled across chapters, for 356 novels. **d**, Share of the variance of the moving-window arc captured by the polynomial arc, by the highest degree kept (median and interquartile range). **e**, The two leading modes of variation among the novels' window arcs, with the Legendre polynomials P_1 and P_2.

Human and machine writing

Readouts that are cheap and do not grow with length can be pointed at a current question: does text written by a language model carry feeling differently from text that people write on the same prompt? Earlier comparisons mostly report that model output is less negative, and more neutral or positive, than human text[54–57], with exceptions by domain[58,59], alongside differences in vocabulary and grammar[60,61]. Ten measures read from signatures and five predictions were fixed before any machine text was generated or scored (Methods): machine text would be more positive overall (HL1), use fewer negative words (HL2), spread its charge more evenly over sentences (HL3) and close on a more positive note than its body (HL4), and a logistic regression on the ten measures would tell it from human text better than chance (HL5).

Two comparisons were made. HC3 pairs questions with an answer from a person and an answer from ChatGPT[54]; 20,091 pairs from five sources had answers long enough to score. For essays, 160 essays written by US school students on eight assignments[62] were sampled, and four Claude models wrote to each student's assignment and length, once on the bare assignment and once with the instruction to write as a student of that grade would. The models ran as agents inside a coding assistant and were not blind to the project (Methods).

In HC3, pooled over its sources, all four directional predictions (HL1 to HL4) hold by the planned rule, and all four effects are small: in human standard deviations, ChatGPT answers differ from human answers by +0.14 in net density, −0.06 in dolor density, −0.04 in clustering and +0.13 in ending lift (Fig. 8a). Source by source, in a breakdown added after review, the verdicts differ (Extended Data Table 6). All four hold in one of the five sources. In the medical questions three of them reverse (net density −0.74, dolor density +0.28, ending lift −0.55), and in the Reddit questions, 70% of the pairs, clustering goes the other way (+0.03).

In the essays one prediction holds under both instructions. Ending lift, the fitted arc of net charge averaged over the last fifth of a text minus the text's density, is 16 per thousand words in the machine essays and 3 in the students', a difference of +0.60 [+0.13, +1.07] standard deviations (+0.52 [+0.16, +0.88] when the model imitates a student). The intervals here and below come from a jackknife over the eight assignments, which is wider than the bootstrap over assignments that the plan named (Methods). The fitted arc weighs the very last words most. A check added after review takes the last fifth literally, as its density less the text's; the difference is then in the same direction and less secure: 14 against 2 per thousand words, +0.51 [−0.01, +1.03] standard deviations, and +0.43 [+0.06, +0.80] as a student. Overall positivity is undecided. Plain essays are less positive than the students' (−0.23 [−0.48, +0.03]) and essays written as a student more positive (+0.29 [−0.01, +0.60]); the planned bootstrap would have called the first a refutation of HL1 and the second support for it. HL2 and HL3 are undecided for essays.

The clearest difference in the essays was not predicted. Machine essays put less of their charge directly after an intensifier (−0.58 [−0.90, −0.25] plain, −0.48 [−0.87, −0.09] as a student). The list of intensifiers includes *so*, *too*, *such* and *most*, which students often use as conjunctions and quantifiers; with those four removed the difference remains (−0.52 [−0.72, −0.32] and −0.32 [−0.60, −0.05]; 1.0% of the students' charge follows one of the remaining intensifiers, against

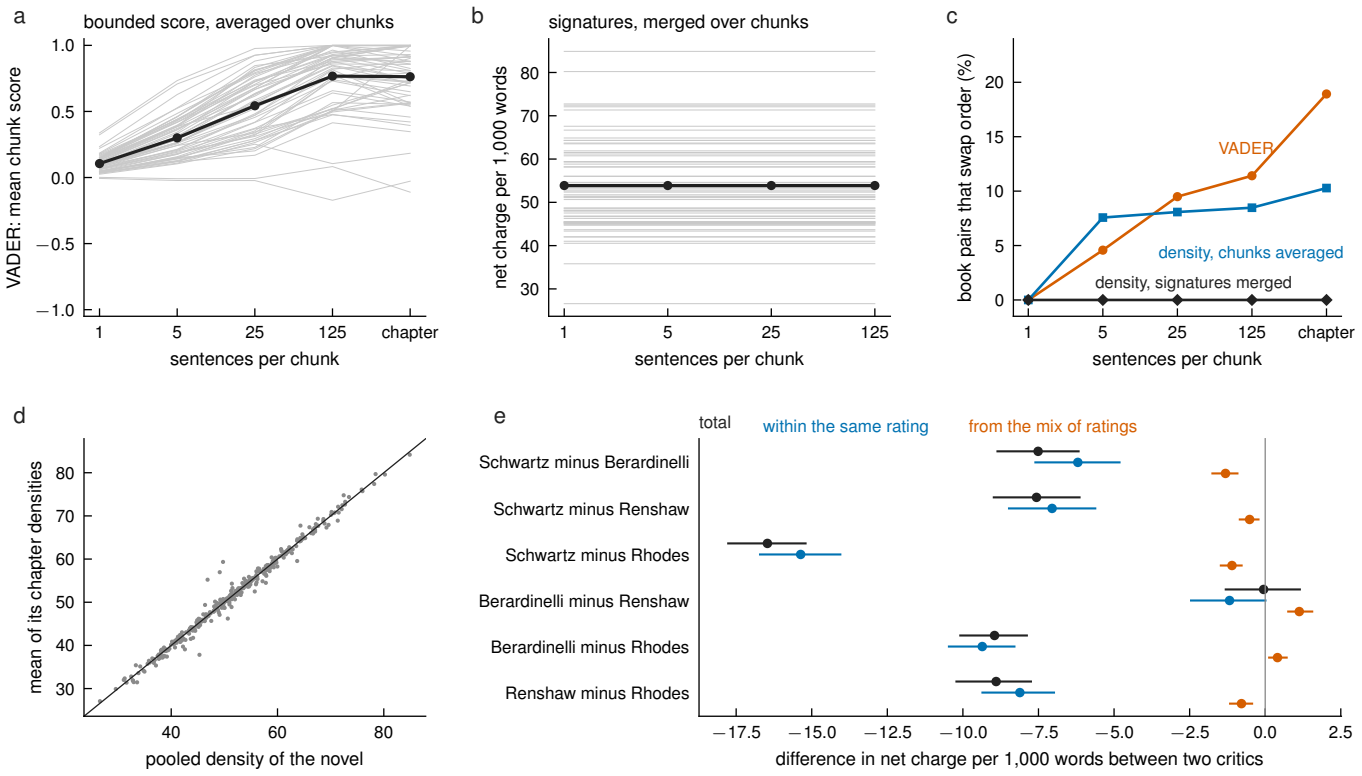


Figure 7 Averaging depends on the cut; merging does not. a, Book score for 60 novels computed as the mean VADER compound score over chunks of a given number of sentences (grey, each novel; black, mean). **b**, Density of the same novels from signatures of the same chunks, merged. **c**, Share of pairs of novels whose order changes relative to one-sentence chunks. **d**, Pooled density of each of 356 novels against the mean of its chapter densities (per 1,000 words). **e**, Differences in density between four film critics, split into the part within the same rating and the part due to the mix of ratings (95% intervals).

0.2% and 0.5% for the models). Pooled HC3 agrees in sign (−0.21), though not in two of its five sources. The finding is exploratory: it is one of six measures for which no prediction was made. Charge under a negation is lower in plain essays (−0.40 [−0.73, −0.06]) and in pooled HC3 (−0.26), and undecided for essays written as a student (−0.24 [−0.62, +0.15]).

HL5 holds, modestly: the area under the curve is 0.63 in HC3, 0.74 [0.65, 0.83] for plain essays and 0.64 [0.52, 0.77] for essays written as a student (jackknife intervals; Fig. 8b). Word count alone does better in HC3 (0.72), where human answers are shorter (median 102 words against 182), and is close to chance for the essays (0.54 [0.52, 0.56]), which were matched on length to within a few per cent (medians 388 and 404 words). The regression fitted on HC3 scores 0.67 and 0.65 on the essays, point estimates without intervals, so part of the pattern may carry from ChatGPT answers of 2023 to Claude essays of 2026. These separations are far too weak for a detector, and purpose-built detectors use much richer evidence and still misfire, notably against non-native writers [63,64].

A tone check tested whether the measures can steer a revision. The model that wrote each of 57 plain machine essays revised it under four instructions: a control instruction, to make it read as a student of that grade wrote it; the control instruction plus a report naming each measure on which the draft lay outside the middle 80% of student essays on other assignments; and the control instruction plus one of two published prose skills written to remove the marks of machine writing (Methods). The planned outcome was a text's mean squared standardised distance from the centre of student essays. It measures how close a text is to the average student essay, and real student essays are not close to it: they average 1.16, because they scatter around their own centre, while drafts average 0.66, control revisions 0.53, revisions made with a skill 0.54 and 0.49, and revisions that saw the report 0.41 (Fig. 8c). A smaller distance means a more average essay. On the planned test the report is not distinguishable from the control (difference −0.12 [−0.29, +0.06]), and neither are the skills (+0.01 [−0.09, +0.10] and −0.04 [−0.17, +0.09]). On the secondary outcome, the mean machine score of a classifier trained on other essays, the drafts are at 0.62, the four kinds of revision between 0.53 and 0.56, and the students at 0.38 (Fig. 8d). Every kind of revision moved a little toward the students, and none came close. The report did move the measures it named: it cut the mean ending lift from +0.31 to +0.02 standard deviations, against +0.21 for the control, and left intensifier use where it was (−0.60 to −0.57), because a draft low on intensifiers is still inside the wide human range and was not flagged.

A report of ranges acts only on drafts outside a wide band, and if it worked perfectly it would pull every draft to one centre. In a second experiment, the dial, each of 60 further drafts was assigned the measured profile of one real student essay, drawn at random from other assignments, and was revised twice by its author model with a sheet giving the current and the target value of each measure; a control arm revised the same drafts twice with the plain instruction. The smallest of the four models declined ten of its 15 first-round dial requests and one of the five that reached a second round, so only four of its drafts are among the 48 essays with both arms complete (15 by Opus, 15 by Fable and 14 by Sonnet), and the result rests mainly on three models. The models followed the dial part of the way. Achieved values correlate with assigned targets at a mean of 0.64 over the ten measures in the dial arm and 0.08 in the control arm (difference +0.56 [+0.30, +0.82]; Fig. 8e), at 0.60 to 0.65 for each of the three

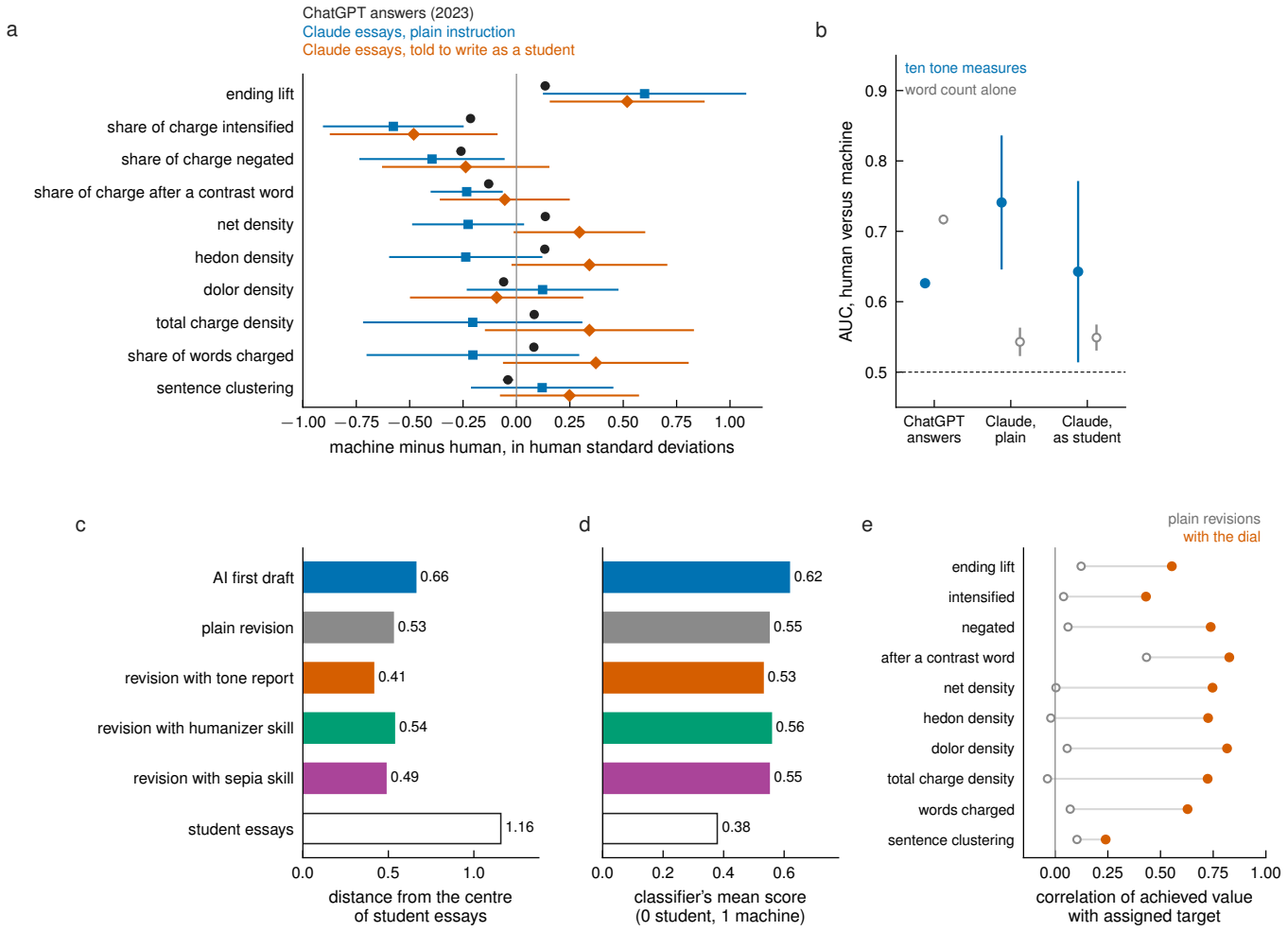


Figure 8 Human and machine writing on the same prompts. a, Difference between machine text and human text on ten measures, in human standard deviations, with 95% intervals: 20,091 questions answered by people and by ChatGPT (bootstrap over questions), and 160 student essays matched by essays from four Claude models under two instructions (158 pairs under the plain instruction, where two requests produced no essay; jackknife over eight assignments). The intervals for HC3 are narrower than the markers. **b**, Area under the curve of a logistic regression that tells human from machine text using the ten measures (blue) or word count alone (grey), scored out of fold, with 95% intervals (bootstrap over questions for HC3, jackknife over assignments for the essays). **c**, Mean squared standardised distance from the centre of student essays for 57 machine drafts, for four kinds of revision of the same drafts, and for the matched student essays themselves, which lie farthest from their own centre. **d**, Mean score of a classifier trained on other essays, from 0 for student essays to 1 for machine essays, for the same texts. **e**, The dial: Pearson correlation across 48 essays between the value assigned to a measure and the value achieved after two revisions, with the dial and with plain revisions.

models with 14 or more essays. By rank correlation the means are 0.58 and 0.09, from 0.78 for dolor density, the best followed, to 0.16 for sentence clustering. The mean squared gap to the target fell from 1.67 to 1.07 after one round and 0.79 after two, about half way, while the control arm stayed at 1.55.

No gain in resemblance to student writing over a plain revision was detected. The classifier's mean machine score is 0.64 for the drafts and 0.43 for the matched student essays; two plain revisions bring it to 0.44 and two rounds with the dial to 0.50 (difference +0.06 [−0.11, +0.23], undecided). The test had little room to show a gain. After two plain revisions the classifier could barely tell the essays from the matched students' (area under the curve 0.52, against 0.70 for the drafts and 0.57 after the dial), so on this measure the plain instruction to write as a student had already done what the dial was meant to do.

A model judge read both revisions of each draft in random order without being told which was which, though it was not blind to the study (Methods). It chose the dial revision as the better written in 30 of 48 pairs (exact 95% interval [47, 76]%). That passes the planned test, that the dial revision not be chosen less than half the time, a test lenient enough that passing it says little. The preference came from two of the writers: 14 of 15 pairs for drafts by Opus and 12 of 15 for Fable, against 3 of 14 for Sonnet and 1 of 4 for Haiku. Asked a second question that the plan did not list, which revision reads more like a student's own work, the judge chose the dial revision in 5 of 48 pairs ([3, 23]%), and never on both questions. That answer is confounded with spelling: the plain revisions wrote informal spellings such as *alot* and *dont*, 4.5 per thousand words against 1.9 in the students' own essays, and the dial revisions wrote none. Extended Data Table 8 shows the essay at the median improvement, chosen by rule. To raise its positive charge the dial revision closes on *a fun, wonderful summer*, and its ending lift moved away from the target.

The measures therefore give a model a partial control over how its text carries feeling, as far as a target can be reached by word choice in two passes. That is narrower than writing as a person does, and a plain instruction to write as a student did as much by the classifier.

The models sometimes refused these requests. Of 17 requests that left no text, 14 were refusals by the smallest model: three in the tone check and 11 in the dial arm, ten of them in its first round. The refusals said that tuning machine text to pass as a student's work, or to move a scoring tool's numbers, misrepresents authorship. The same model

followed the plain instruction to write as a student in 44 of its 45 plain requests (all 30 in the dial and 14 of 15 in the tone check), and declined two of its 15 revisions with the report and 11 of its 20 dial requests, so it was mainly the steering sheets that it declined. The remaining requests failed for other reasons (Methods). The tools released with this paper report tone against a human range, for writers who disclose their use of a model. On these measures the plain instruction to write as a student did the most to make machine essays hard to tell from students', and the measures are too weak to serve as a detector.

Discussion

The aim was a measure of sentiment for which granularity holds. The felicific signature provides one: charge adds, position sums shift, and the signature of a whole is the merge of the signatures of its parts, exactly in integer arithmetic and to rounding error in floating point. Totals, density, a level with a standard error, a predicted rating and an arc are all read from the same 19 numbers, and the scale at which to read them can be chosen after the text has been scanned.

Its components are established: averaged word norms [2], separate positive and negative counts [3,9], mergeable moment summaries [20,21] and the decompositions of pooled values [51,52] have long histories in their own fields, and the description of mergeable position weights is a classical result [30,31]. The paper adds their combination for sentiment, an implementation tested for exactness, and evidence on how position and neutral words enter human judgements.

Of the three hypotheses about how sentiment is distributed in text that were specified in advance, front-loading and recency were not supported and dilution was. Feeling words are not placed early, and whether early words weigh more with readers depends on the genre: they do in film-review sentences, and the reverse holds in tweets and in Yelp reviews. Recency met its planned tests, but the two that concern readers compared parts of unequal length and reverse once length is held fixed; a test fixed in advance prevents choosing the analysis after seeing the result, and it must still control for confounds such as length. Neutral words do dilute, as the averaging account of impression formation predicts [34,35], and in exploratory analyses of phrase ratings they do so from either side. What the phrase ratings support is plainer than either hypothesis: a part's weight in the whole grows with its share of the words, by less than proportion.

The signature is not the most accurate reader of a single text. Transformers agree with raters better on average (0.59 to 0.68 against 0.51), and where the judgement of one text matters they are the better tool. On long documents it is weaker still: on whole film reviews a plain lexicon method beat it, and wherever validity above the sentence could be checked, an average of sentence scores from another method tracked the human target better than the merged signature did. The case for the signature rests on other properties: thousands of times the throughput per processor core, a result that does not change with where a text is cut between sentences, and readouts that can be audited word by word. One sensible division of labour is to scan everything with signatures and to send a larger model only the spans that the scan marks.

Several limits qualify these results. Word charges ignore events, as the last chapter of *Moby-Dick* shows, and irony, sarcasm and meaning carried by syntax beyond the four operators are missed. A word has one charge whatever its sense or part of speech: *like*, *kind* and *pretty* are charged in every use. The norms also carry their raters' attitudes. *Deaf*, *disabled* and *blind* are rated about half a dolor and *old* a quarter, so a text that mentions disability or age scores as more negative for that alone, a known fault of sentiment lexicons and of systems built on them [65,66]; charges refined on rated text made this worse. The lexicon is English and its ratings date from the early 2010s, and 69% of its charged entries come from VADER's list, so the comparison with VADER is between related lexicons. The constants were fitted on six corpora in which reviews and social media dominate, and their readings change between corpora. Film reviews were scored with credits and sign-off lines left in. No human ratings of chapters or books were available, and the rule that finds chapters is approximate. The standard error of a level treats words as independent draws, which the clustering of charge in sentences and chapters contradicts, so the section intervals in Fig. 6a are too narrow. Human judgements of long experiences are known to weigh peaks and endings rather than sums [67]; the signature keeps what an averaging or an end-weighted readout needs, and a readout of peaks would need signatures kept per part. The comparison of human and machine writing covers 160 student essays on eight assignments, four models from one family reached through an agent harness that did not keep them blind, and ChatGPT answers from 2023; the dial rests mainly on three of those models, and its judge was a model. Analyses that were not in the plan are listed in Methods (Pre-specification) and marked where they are reported.

Three additions would test the measure further: human ratings of chapters, so that readouts above the sentence can be validated directly; word norms in other languages; and the decayed sum that Proposition 2 allows, for a running level in streams of speech or messages.

Methods

Word charges. labMT 1.0 gives the mean of 50 ratings of happiness on a scale from 1 to 9 for 10,222 frequent English words [2]. The charge of a word is $c = (h - 5)/4$, so that $c = 1$ at the top of the scale, 0 at its neutral point and -1 at the bottom. The VADER list gives the mean of 10 ratings from -4 to 4 for 7,494 words, emoticons and abbreviations [7]. It was put on the labMT scale with one slope through the origin, the ratio of root-mean-square values over the 1,366 shared words, and the two values were averaged where both exist. The result has 16,350 entries, of which 8,408 have $|c| \geq 0.25$. Operator words are given no charge, which leaves 8,370 entries that carry charge; the rest are buffer words, as are words outside the list. Positive charge is $h = \max(c, 0)$ hedons and negative charge $d = \max(-c, 0)$ dolors. Charges are stored as integers in thousandths. Adding the NRC-VAD [68] and Warriner [69] norms ($r = 0.88$ and 0.92 with labMT on shared words) raised mean agreement on development data from 0.521 to 0.534; their licences do not allow redistribution, so they are not part of the shipped lexicon and no reported result uses them.

Tokens and operators. Text is lower-cased. Emoji are replaced by their names, using the list distributed with VADER, and 40 emoticons are recognised when they follow a space. Words are runs of letters and apostrophes. A full stop, exclamation mark or question mark ends a sentence; a comma, semicolon or colon ends a clause. Four closed lists of operator words carry no charge themselves: negators (*not*, *never*, forms ending in *n't* and others), twenty intensifiers (*very*, *really*, *extremely*, *absolutely*, *completely*, *totally*, *utterly*, *incredibly*, *so*, *too*, *highly*, *especially*, *particularly*, *truly*, *quite*, *super*, *such*, *most*, *deeply*, *thoroughly*), diminishers (*slightly*, *somewhat* and others) and contrast words (*but*, *however*, *yet*, *nevertheless*, *nonetheless*). A negator reaches the next three words or the next punctuation mark, whichever comes first; an intensifier or diminisher reaches the next two words or the next punctuation mark, whichever comes first; a contrast word reaches the end of its sentence. Every sentence end clears all operators. The scan is a finite-state transducer and uses string replacement, translation and splitting only.

Signature. For a span of L words with charges h_i, d_i at positions $i = 0, \ldots, L-1$, the signature holds L; the number E of charged words; $S_k^{\pm}$ for $k = 0, \ldots, 3$ (equation 1); the hedons and the dolors inside the reach of each of the four operators; and $\sum_i (h_i - d_i)^2$. The fast implementation computes signatures for a batch of texts with array operations. A reference implementation loops over tokens in integer arithmetic and defines the result; the test suite checks that the two agree, and checks Proposition 1, the two-row prefix query and merging at cuts inside a sentence for exact equality on random texts.

Proposition 1. Let A and B be texts, joined by white space, and let A end at a sentence end. Then $\sigma(AB) = \sigma(A) \oplus \sigma(B)$, where $\oplus$ adds every entry except the position sums and applies equation (2) to those. The operation $\oplus$ is associative, the zero vector is its identity, and for fixed $\sigma(A)$ the map $\sigma(B) \mapsto \sigma(A) \oplus \sigma(B)$ is invertible.

Proof. Operators do not reach across a sentence end, so each word of B has the same charge and the same operator context in AB as in B. Counts and totals therefore add. A word at position j of B is at position $L_A + j$ of AB, and expanding $(L_A + j)^k$ gives equation (2). Write T_ℓ for the linear map that equation (2) applies to the position sums for a shift by ℓ. It is the matrix of the substitution $x \mapsto x + \ell$ on polynomials of degree below four in the monomial basis, so $T_\ell T_m = T_{\ell+m}$ and $T_\ell^{-1} = T_{-\ell}$, which give associativity and invertibility. □

Merging is not commutative: parts can be scored in any order and on any machine, and must be merged in reading order. With prefix signatures $\sigma_j = \sigma(s_1 \cdots s_j)$ over sentences, the signature of sentences i to j is $T_{-L}(\sigma_j - \sigma_{i-1})$ in the position sums, with L the length of the first $i-1$ sentences. In floating point this difference keeps the counts and totals and loses the higher position sums of short spans deep in long texts: for one sentence at the end of 200,000 words the cubic sum can be wrong by more than its own size. The integer implementation is exact, and in floats the position sums of such a span are obtained by merging its own rows forward, which is accurate at any depth.

Proposition 2. Let $\varphi_1, \ldots, \varphi_n$ be linearly independent real sequences on positions $0, 1, 2, \ldots$, and for a real charge sequence c of length L let $\Phi(c) = \big(L, \sum_i c_i\varphi_1(i), \ldots, \sum_i c_i\varphi_n(i)\big)$. There is a function F with $\Phi(cc') = F(\Phi(c), \Phi(c'))$ for all charge sequences c, c' if and only if the span V of the φ_r is closed under the shift $(T\varphi)(i) = \varphi(i+1)$. In that case every φ_r satisfies a linear recurrence with constant coefficients of order at most n, and is a combination of sequences $i^k\rho^i$, with ρ a nonzero complex number, and of sequences that vanish from position n onward.

Proof. If V is closed under the shift, then $T^\ell\varphi_r = \sum_s M^{(\ell)}_{rs}\varphi_s$ for some matrices $M^{(\ell)}$, and $\sum_j c'_j\varphi_r(L+j) = \sum_s M^{(L)}_{rs}\sum_j c'_j\varphi_s(j)$, so $\Phi(cc')$ follows from $\Phi(c)$ and $\Phi(c')$. Conversely, fix $\ell \geq 1$ and let c be ℓ zeros. Take J large enough that the restrictions of the φ_s to positions below J are linearly independent, and let c' range over sequences of length J. By assumption $\sum_j c'_j\varphi_r(\ell+j)$ depends on c' only through the numbers $\sum_j c'_j\varphi_s(j)$. Adding to c' any d with $\sum_j d_j\varphi_s(j) = 0$ for all s therefore leaves it unchanged, so the linear functional $d \mapsto \sum_j d_j\varphi_r(\ell+j)$ vanishes on the common kernel of the n functionals $d \mapsto \sum_j d_j\varphi_s(j)$ and is a linear combination of them. The coefficients are unique, and do not change with J because the equations for J are among those for $J+1$. Hence $T^\ell\varphi_r \in V$. The shift is then a linear map on the n-dimensional space V and, by the Cayley–Hamilton theorem, is annihilated by its characteristic polynomial p, so every member of V satisfies the recurrence $p(T)\varphi = 0$. Its solutions are spanned by $i^k\rho^i$ for the nonzero roots ρ of p and, for a zero root of multiplicity m, by the sequences supported on positions below m. □

The proposition concerns summaries that are linear in the charges. It is the sequence form of the description of finite-dimensional translation-invariant function spaces [30,31], and of the fact that a function on strings has a finite-dimensional linear representation exactly when its Hankel matrix has finite rank [32,33]. A geometric weight ρ^i gives the one-number case $S_\rho(AB) = S_\rho(A) + \rho^{L_A}S_\rho(B)$. On relative position: suppose a continuous function g on $[0,1]$ is available from one summary for every length, meaning that for each L the sequence $g\big((i+\frac{1}{2})/L\big)$, $0 \leq i < L$, agrees on those positions with a member of a fixed shift-closed space V of dimension n. Then g is a polynomial of degree below n. Let p be the characteristic polynomial of the shift on V and write $p(z) = (z-1)^m q(z)$ with $q(1) \neq 0$. For a smooth test function ψ supported inside $(0,1)$ and L large, summing $\psi(i/L)$ against $p(T)$ applied to the sequence gives zero, and summation by parts moves the operator onto ψ: $\sum_j g\big((j+\frac{1}{2})/L\big)\,[p(T^{-1})\psi_L](j) = 0$ with $\psi_L(j) = \psi(j/L)$. Since $(T^{-1}-1)^m\psi_L = (-1)^m L^{-m}\psi^{(m)}(j/L) + O(L^{-m-1})$ and $q(T^{-1})\psi_L = q(1)\psi_L + O(L^{-1})$, multiplying by L^{m-1} and letting L grow turns the sum into $(-1)^m q(1)\int_0^1 g\,\psi^{(m)} = 0$. So the mth derivative of g vanishes as a distribution, and g is a polynomial of degree below $m \leq n$. Continuity is needed: the points $(i+\frac{1}{2})/L$ for odd L and for even L form disjoint sets, so the function equal to 1 at the points of odd lengths and 0 elsewhere gives the total charge for odd L and 0 for even L; the summary (L, S_0) supplies it, although it is not a polynomial. Conversely the signature supplies every polynomial weight of degree three or less.

Readouts. Density is $1000\,(H-D)/L$. Level is equation (3), a ratio $R = (H-D)/(a+E+\kappa B)$ whose numerator and denominator both vary from word to word: a word adds its charge $c_i = h_i - d_i$ to the first and w_i to the second, with $w_i = 1$ for a charged word and κ for a buffer word. Treating words as independent draws, the delta method gives the standard error $1000\sqrt{\sum_i u_i^2 - (Ra)^2}/L\,/\,(a+E+\kappa B)$ with $u_i = c_i - Rw_i$ and $\sum_i u_i^2 = \sum_i c_i^2 - 2R(H-D) + R^2(E+\kappa^2 B)$, all read from the signature. For density ($a=0$, $\kappa=1$) it is the standard error of a mean, with the variance of the charges taken over L rather than $L-1$. In simulated texts of independent words it matches the spread of the level to within about 4% at 200 words and 5% at 50; being a first-order approximation, it is about 10% too small at 20 words and about 20% too small at 10. With $x_i = (2i+1)/L - 1$ and Legendre polynomials P_k, the moments $m_k = \sum_i (h_i - d_i)P_k(x_i)$, $k \leq 3$, are linear combinations of the position sums, and the arc is $f(x) = \sum_k \frac{2k+1}{L} m_k P_k(x)$. The sampled polynomials are orthogonal up to terms of order L^{-2}, so f is the least-squares cubic to that order: in random texts the largest difference between the two is about 7% of the cubic's range (largest minus smallest value) at 10 words and never more than 12% there, about 2% at 20 words and 1% at 30, and the two agree to one part in 10^5 over the novels. Below ten words the arc is a poor approximation (about 30% of the range apart at five words), and below four words no cubic is determined; the implementation then returns no arc. The calibrated valence is $b_0 + \mathbf{b}\cdot\mathbf{x}/(a+E+\kappa B)$, where $\mathbf{x}$ holds H, D, the moments m_1, m_2, m_3, the negated hedons, the negated dolors, and the net charge that is intensified, diminished, and after a contrast word.

Corpora. Table 1 lists the rated corpora. Official training, development and test portions were kept where they exist. Facebook posts and film reviews were split 40/10/50 by a seeded shuffle. Yelp training and development portions are seeded samples of the official training set.

Table 1 Rated corpora. Numbers of texts in the training, development and test portions.

Corpus	Unit	Human rating	Train / dev / test
EmoBank [37]	sentence, six genres	valence, 1 to 5	8,062 / 1,000 / 1,000
SemEval-2018 [38]	tweet	valence, 0 to 1	1,181 / 449 / 937
Facebook posts [39]	post	valence, 1 to 9	1,158 / 289 / 1,448
SST [40]	sentence	sentiment, 0 to 1	8,544 / 1,101 / 2,210
Film reviews [41]	review	critic's rating	2,002 / 501 / 2,503
Yelp [42]	review	stars, 1 to 5	100,000 / 10,000 / 50,000

Fitting and evaluation. Ratings were standardised within each corpus with the mean and standard deviation of its training portion, and corpora were given equal total weight. For each target corpus and each pair (a, κ) on the grid $a \in \{0, 0.5, 1, 2, 4, 8, 16, 32\}$, $\kappa \in \{0, 0.05, 0.1, 0.2, 0.5, 1, 2\}$, the coefficients were fitted by ridge regression on the training portions of the other five corpora (penalty 10^{-3} times the mean diagonal of the weighted Gram matrix; intercept unpenalised). The pair with the highest mean Pearson r on those corpora's development portions was kept, the coefficients were refitted on training plus development portions, and the target's test portion was scored once. The shipped constants ($a=4$, $\kappa=0.5$) are the same fit on all six corpora. Intervals are 95% percentile bootstrap intervals, drawing source documents for EmoBank and items elsewhere; differences between methods use the same resamples. Pearson r and differences between methods use 2,000 resamples, as planned. To bound running time Spearman's ρ uses 500, and the position, dilution, treebank and machine-writing analyses use between 100 and 2,000; the scripts give each count. The interval for the mean difference from VADER is a t interval over the five primary corpora. Comparators: VADER's compound score (version 3.3.2) for the whole text, as planned, and, added after review, its mean over the sentences of the text; the hedonometer, the mean labMT value of words outside its neutral band [2]; the polarity score of the TextBlob library (version 0.20.1); ridge regression on TF-IDF weights of words and word pairs, trained on the target's own training portion or on the other five corpora; and two transformer classifiers scored as $P(\text{positive}) - P(\text{negative})$ on text truncated to 512 tokens, DistilBERT fine-tuned on SST-2 [12] and a RoBERTa model fine-tuned on tweets [44]. Both were used as distributed, with no tuning to these corpora, and text was passed to them unchanged, without the replacement of user handles that the RoBERTa model's documentation prescribes for tweets. The transformers were run on a seeded sample of 5,000 Yelp test reviews, and for Yelp all methods are compared on that sample.

Refined charges. With exposure $a+L$ and operator multipliers θ from a one-channel fit, the predicted rating of text t is $b_0 + s\sum_i \mu_i c_{w_i}/(a+L_t)$ with $\mu_i = 1 + \theta\cdot(\text{operator flags of word } i)$, which is linear in the vector c of word charges. The refined charges minimise the weighted squared error plus $\lambda\|s(c-c^0)\|^2$, where c^0 holds the norm charges and words that occur in at least 25 training texts start at zero. The penalty $\lambda \in \{0.03, 0.1, 0.3\}$ was chosen on development data of the fitting corpora, and the readout was then refitted with the refined charges. Each step is a ridge regression; the sequence is not a joint optimum.

Position and dilution. The plan labels front-loading H1, with predictions H1a to H1c, and dilution H2. For where charge sits (planned, H1a), each text of at least ten words contributes $\sum_i |c_i|x_i$ and $\sum_i |c_i|$, and the statistic is the ratio of their sums over texts. Besides the six rated corpora, the units were paragraphs of the film reviews and chapters of 150 of the novels, so the eight kinds of unit come from seven bodies of text. The position kernel of a corpus (planned, H1b) is $1 + (b_1P_1 + b_2P_2 + b_3P_3)/b_0$ from a one-channel fit to all its portions with the shipped a and κ; the contrast is its mean over the first third minus its mean over the last third. For EmoBank documents of eight or more sentences (H1c, which the plan marks exploratory), $|V-3|$ was regressed on relative sentence position with document means removed, resampling documents. The planned test of dilution (H2) is the paired difference in held-out r between the calibrated readout and the same readout refitted with $\kappa = 0$.

The remaining analyses were not in the plan. For dilution in the corpora, rated intensity is the absolute deviation of a rating from the corpus median, and the partial rank correlation with the number of buffer words removes the rank of total charge from both by linear regression. The curves of agreement against κ are fits to single corpora with a fixed at the shipped value, on a grid extended to $\kappa = 4$. For the treebank, every binary node whose phrase and both parts have ratings in the phrase dictionary was used (100% of nodes in the training trees); ratings were centred at 0.5, the phrase rating was regressed on the two part ratings, and sentences were resampled. With length held fixed, nodes whose parts both have two or more words were split by which part is longer, and the weights were also fitted as one step function of a part's share of the phrase's words in ten bins, shared by both parts, plus a constant for the later part; a second fit gives each part its own step function (Fig. 4b). For dilution in the treebank, a part counts as rated if its centred rating is at least 0.15 in absolute value and as neutral if at most 0.05; the share kept is the ratio of sums of the signed phrase rating to the absolute part rating. The matched comparison of sides weights cells defined by the length of the rated part and the length of the neutral part (1, 2, 3 to 4, 5 to 8, 9 or more words each) by the smaller of their two counts. These analyses are reported for all trees, with the development and test trees alone where the recency tests are discussed.

Recency tests. The second addendum to the plan fixed three tests. R1: in binary nodes of the treebank's development and test trees with both parts of two or more words, the weight of the later part minus that of the earlier part in the regression of the phrase rating on the two part ratings. R2: in the same trees, the share of its rating that a rated part keeps when two or more neutral words come before it, minus the share when they follow it. R3:

in the 22,001 reviews of 100 or more words in the test portion of the IMDB corpus [45], the area under the curve for telling favourable from unfavourable reviews by net charge per word of the last fifth, minus the same for the first fifth. Each was to count as support if its 95% bootstrap interval lay above zero (2,000 resamples of sentences for R1 and R2, 1,000 of reviews for R3). All three did. The controls for length described above were not part of the addendum.

Novels, fairy tales and critics. From a seeded sample of English fiction in the Project Gutenberg catalogue, 640 texts were fetched from a mirror, and 356 had at least ten chapters of 300 or more words and 20,000 words in all after removal of front matter. Chapters were found by headings, sections under 300 words were dropped, and each chapter was closed with a full stop. The rule is approximate. In *Moby-Dick* it finds 129 sections for the book's 135 chapters and epilogue: it drops the chapters shorter than 300 words, joins the epilogue to the last chapter, and keeps one block of transcriber's notes as a section. The null for chapter differences shuffles a novel's sentences 200 times and keeps the number of sentences in each chapter, so the smallest attainable p is $1/201$. The comparison of chunk sizes with VADER used the first 60 novels in file order. Arcs were compared on 100 points; the moving window is 10% of the words. The fairy-tale annotations [50] give two annotators' emotion labels for each sentence; a label of happy or positively surprised was scored +1, neutral 0, and any other emotion −1, and the two were averaged. Critic comparisons use all 5,006 reviews with the four-level rating as stratum; intervals resample reviews within critic. The film reviews were scored as distributed. Credits lines, reviewers' sign-off lines and footers were not removed; they sit at the start and the end of a review, where position is measured, and one critic's standard footer, an invitation to visit his web page, closes 486 of his 902 reviews and adds 0.65 per thousand words to his pooled density.

Human and machine writing. The ten measures are: hedons, dolors, net charge and total charge per thousand words; the percentage of words that are charged; the percentage of charge that is negated, intensified, and after a contrast word; ending lift, the mean of the cubic arc over the last fifth of the text minus the text's density; and sentence clustering, the between-sentence variance of density, weighted by words, divided by its expected value if the text's words were dealt to its sentences at random, $\sigma^2(m-1)/L$ for m sentences, L words and word-level variance σ^2 of net charge. None grows with the length of a text, though all are noisier in short ones. Two have caveats. The cubic arc weighs words up to the end of the text, the last ones most, so ending lift is not the density of the last fifth; that literal quantity is reported beside it. Clustering is 1 under the null for sentences of any length, but where charge is correlated within sentences it grows with sentence length, so a difference in clustering can reflect a difference in sentence length. From HC3 [54] the first human answer and the first ChatGPT answer to each question were taken, keeping the 20,091 questions where both have at least 30 words; contractions split by the corpus's tokenisation of human answers were rejoined. From PERSUADE 2.0 [62], 20 essays of 150 to 800 words were drawn for each of the eight assignments that need no source text (160 essays, seed 0). Each essay was matched by two machine essays written to the same assignment and word count, one with no further instruction and one with the instruction to write as a student of the same grade would in a timed test. Two of the 320 requests produced no essay, both under the plain instruction: the agents set the task aside as unrelated to the message the harness had relayed to them (below).

The writers were subagents of the coding assistant, one essay per agent, running Claude Fable 5.2, Opus 5.5, Sonnet 5.5 and Haiku 5.5 in rotation at low reasoning effort. Each agent's prompt held its task and a file path to write to. The harness also attached its standing context to every agent: an index of the author's project notes, a coding-style plug-in and the author's most recent message to the assistant. For the essays that message described the planned comparison of machine and human essays; for the dial revisions and the judges it stated the author's hope that steering by these measures would make machine text read as more human. The file paths named the experimental arm. The agents were therefore not blind to the project, and the revisers and the judge were not blind to the hoped-for result. The job files and the workflow scripts that dispatched them, with the judge's instruction, are in the repository, as is the final message of each of the 17 requests that left no text (`machine_text/declined_requests.json`); full transcripts are not.

Differences are paired by essay or question and standardised by the human standard deviation. HC3 intervals are percentile bootstrap intervals over questions. For the essays the plan named a bootstrap over the eight assignments. A percentile bootstrap over eight clusters is too narrow, so the printed intervals are from a delete-one-assignment jackknife with a t reference on seven degrees of freedom (six for the tone check and the dial, which cover seven assignments), and the planned verdicts are stated where they differ. A bootstrap over essays, which is neither the planned rule nor the printed one, would call HL3 refuted for essays written as a student, which cluster more than the students' essays. The classifier is a logistic regression on the ten standardised measures, scored out of fold with all texts of an assignment or question in one fold; its intervals are bootstrap intervals over questions for HC3 and jackknife intervals over assignments for the essays. The check on intensifiers recomputes that measure with *so*, *too*, *such* and *most* taken off the list of intensifiers.

For the tone check, 60 plain-instruction drafts, 15 per model drawn with one seed, were each revised by the model that wrote them under four instructions: the control instruction, to revise the draft so that it reads as a student of that grade wrote it; the control instruction plus a report listing each measure on which the draft lay outside the middle 80% of student essays on other assignments; and the control instruction plus an instruction to do the rewrite with one of two published prose skills for removing the marks of machine writing (*humanizer*, built on Wikipedia's page on signs of AI writing, and *sepia-recreate*; both are public plug-ins for the coding assistant). Distance is the mean over measures of the squared standardised difference from the mean of student essays on other assignments. The smallest model declined three of the 240 revision requests, stating that making machine text read as a student's work would misrepresent authorship; essays with all four revisions (57, from seven of the eight assignments) enter the comparison. The dial used 60 further drafts drawn the same way; its 48 complete pairs come from seven assignments in clusters of 15, 13, 8, 4, 3, 3, 2 essays. One control revision was reported as written and left no file. The dial's classifier was trained on the 98 essay pairs outside the dial sample. The judge was Claude Opus 5.5, or Sonnet 5.5 for drafts written by Opus; shares chosen by the judge are given with exact binomial intervals that treat pairs as independent, which the split by writer shows they are not. The judge saw no real student essay, and the key to the arms was a file in the same folder as its job files. Informal spellings are counted from a fixed list: *alot* and seventeen contractions written without their apostrophe.

Throughput. Timings were taken on a rented machine (AMD Ryzen 5 5600X 6-Core Processor, six cores and twelve threads; RTX 3090) so that processor and graphics-card figures come from one host. The plan had named the author's laptop, which was replaced because long runs overheated it. The texts are 200,000 sentences from Yelp reviews (mean 16 words). Each method scored a prefix of the same list: all of it for the signature, 20,000 sentences for the hedonometer, 5,000 for VADER and TextBlob, 600 for each transformer on one thread, 3,000 on twelve threads and 20,000 on the graphics card (batches of 128 sorted by length, single and half precision). The timed operation for the signature is the scan to signatures followed by the level readout; the calibrated readout takes about 6% longer, measured afterwards on the author's laptop. Signature timings are the best of three runs and lexicon timings the best of two; each transformer was timed once.

Pre-specification. The method, the corpora, the outcomes, the comparators and the decision rules for the hypotheses were written into an analysis plan and committed to the project's git repository (commit `b7b6f9b`) before any evaluation data were scored. Four addenda were committed before the data they concern were generated or examined: the comparison of human and machine writing (`15eec69`), the recency hypothesis and its three tests (`2a2aa34`), the rewrites with prose skills (`966edc2`) and the dial (`4c53156`). This is a record kept by the author and not an independent registration. The commits were made in a local repository over one evening, they carry the timestamps of the author's machine, and they were pushed to the public repository only when the study was complete. The repository's history was later rewritten to remove credentials and a Windows executable left from 2024, a refined lexicon that is not part of the release, and absolute local paths; file contents are otherwise unchanged, as are all commit dates, and the identifiers given here are those after the rewrite. Some of these commit messages say 'pre-register'. The order of file times agrees with the stated order, with one exception: the note that narrows the essay sample to eight assignments was committed together with the second addendum, after the essays had been written.

Test portions were first scored with the plan as written; the results of that first run were not kept, so whether the accuracy criterion was met before the changes below cannot be checked from the repository. Deviations, all listed in the plan: the grid for a and κ was widened after the first run chose its upper edge, without changing the choice; the tokeniser was rewritten for speed; a bug in the selection of constants for the uncalibrated level was fixed; the VADER list was put on the labMT scale by a ratio of root-mean-square values where the plan said regression; several bootstraps use fewer resamples than the planned 2,000; the essay intervals are jackknife intervals; the planned report of how often the document score and the mean of sentence scores differ in sign was at first made for the transformers only, and VADER was added later; and the classifier for the dial was first trained on a set that left out only the essays with both arms complete, then retrained leaving out all 60 sampled essays. Refined charges, the bag-of-words ridge trained on other corpora, the partial correlations, the curves against κ, every treebank analysis other than the recency tests R1 and R2, the readings of the constants, the check of error against exposure and the second question to the judge were added after the first results and are exploratory. The length-controlled treebank analysis, the jackknife intervals, the table of HC3 by source, the check of refined charges for words about groups, sentence-averaged VADER, the corrected standard error of the level, the literal last-fifth check, the check on ambiguous intensifiers and the count of informal spellings were added in response to reports by referee agents. Where a model declined a request, the text is missing and the essay leaves the paired comparison; no declined request was retried. A February 2026 version of the method, built on word embeddings, failed its own criterion of matching VADER on GoEmotions [70]; the criterion carried over here is a mean r over the primary corpora at least that of VADER.

Software. Python 3.10, NumPy 2.0, SciPy 1.15, scikit-learn 1.5, PyTorch 2.11 and Transformers 4.51 on the author's machine; the timings used PyTorch 2.3.1 on the rented host. Figures were drawn with Matplotlib 3.9 and SciencePlots [71]; the overview diagram was drawn with the FigCraft command-line tool from `scripts/make_overview.py`.

Use of AI tools. The aims and the two hypotheses come from the author's unpublished 2024 draft. The code, the analyses and the text of this article were produced with Claude (Anthropic), an AI assistant working in a coding environment under the author's direction, and were reviewed by AI referee agents whose reports are in the repository under `reviews/`. One of those reports found the confound that led to the withdrawal of the recency hypothesis. The essays in the comparison of human and machine writing were written by Claude models by design. Responsibility for the content rests with the author.

Data availability

All corpora and lexicons are public and are downloaded by `scripts/fetch_data.py`. The repository redistributes the lexicon built from labMT (CC BY) and the VADER word list (MIT), VADER's emoji list, summary statistics of the reference texts, the machine-written essays, four complete student essays from PERSUADE 2.0 (CC BY 4.0), quoted with their identifiers in a comparison page, the result files behind every number and

figure, and the analysis plan; licences and attributions are in `THIRD_PARTY_NOTICES.md`. The folder `legacy/2024` keeps the scripts and data of the author's 2024 draft and is not used by this study.

Code availability

`https://github.com/ayazkhan27/anonymity`. The numbers in the text are generated by `scripts/make_numbers.py` from the result files, the released lexicon and, for the worked example and one count, the downloaded data.

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Competing interests

The author declares no competing interests.

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Extended data

Extended Data Table 1 Pearson r with human ratings on held-out test portions, with 95% bootstrap intervals. The mean is over the five corpora with continuous human ratings (all but Yelp). Signature constants are fitted on the other five corpora unless stated.

	EmoBank	Tweets	Facebook	SST	Film reviews	Yelp	Mean
n (test portion)	1,000	937	1,448	2,210	2,503	5,000	
Signature, calibrated	0.48 [0.43, 0.54]	0.70 [0.67, 0.73]	0.62 [0.58, 0.65]	0.44 [0.40, 0.47]	0.29 [0.26, 0.33]	0.61 [0.59, 0.62]	0.51
Signature, level only	0.44 [0.39, 0.49]	0.71 [0.67, 0.74]	0.57 [0.54, 0.60]	0.42 [0.39, 0.46]	0.26 [0.22, 0.29]	0.54 [0.52, 0.56]	0.48
Signature, constants from the same corpus	0.48 [0.43, 0.55]	0.73 [0.70, 0.76]	0.62 [0.58, 0.65]	0.46 [0.43, 0.49]	0.40 [0.37, 0.44]	0.63 [0.61, 0.64]	0.54
Signature, refined charges	0.41 [0.36, 0.46]	0.71 [0.68, 0.74]	0.58 [0.55, 0.61]	0.55 [0.52, 0.58]	0.49 [0.47, 0.52]	0.64 [0.62, 0.65]	0.55
VADER, whole text	0.46 [0.41, 0.50]	0.71 [0.68, 0.75]	0.58 [0.54, 0.62]	0.45 [0.42, 0.48]	0.25 [0.21, 0.29]	0.55 [0.54, 0.57]	0.49
VADER, mean of sentence scores	0.46 [0.41, 0.51]	0.69 [0.66, 0.73]	0.56 [0.52, 0.60]	0.44 [0.41, 0.48]	0.30 [0.27, 0.34]	0.63 [0.61, 0.64]	0.49
Hedonometer	0.44 [0.40, 0.48]	0.59 [0.55, 0.63]	0.49 [0.45, 0.52]	0.34 [0.30, 0.37]	0.24 [0.20, 0.28]	0.54 [0.52, 0.56]	0.42
TextBlob	0.31 [0.24, 0.36]	0.51 [0.46, 0.56]	0.53 [0.49, 0.57]	0.41 [0.37, 0.44]	0.46 [0.43, 0.49]	0.60 [0.58, 0.61]	0.44
Bag-of-words ridge, other corpora	0.32 [0.27, 0.40]	0.64 [0.60, 0.67]	0.58 [0.54, 0.61]	0.56 [0.53, 0.58]	0.64 [0.61, 0.66]	0.67 [0.65, 0.68]	0.55
Bag-of-words ridge, same corpus	0.51 [0.47, 0.55]	0.59 [0.55, 0.63]	0.53 [0.50, 0.57]	0.66 [0.64, 0.68]	0.77 [0.75, 0.78]	0.83 [0.83, 0.84]	0.61
DistilBERT (SST-2)	0.50 [0.46, 0.53]	0.60 [0.56, 0.64]	0.54 [0.50, 0.57]	0.78 [0.76, 0.79]	0.56 [0.53, 0.58]	0.75 [0.73, 0.76]	0.59
Twitter-RoBERTa	0.73 [0.70, 0.76]	0.81 [0.78, 0.83]	0.78 [0.76, 0.80]	0.77 [0.75, 0.79]	0.31 [0.27, 0.34]	0.76 [0.75, 0.77]	0.68

Extended Data Table 2 Spearman's ρ with human ratings, the plan's secondary outcome, laid out as Extended Data Table 1 (500 resamples).

	EmoBank	Tweets	Facebook	SST	Film reviews	Yelp	Mean
Signature, calibrated	0.48 [0.43, 0.53]	0.71 [0.67, 0.74]	0.60 [0.56, 0.63]	0.44 [0.40, 0.47]	0.30 [0.26, 0.33]	0.64 [0.63, 0.66]	0.51
Signature, level only	0.43 [0.37, 0.48]	0.72 [0.68, 0.74]	0.58 [0.54, 0.61]	0.42 [0.38, 0.45]	0.26 [0.23, 0.30]	0.59 [0.56, 0.60]	0.48
Signature, constants from the same corpus	0.50 [0.45, 0.54]	0.74 [0.70, 0.76]	0.61 [0.57, 0.64]	0.45 [0.41, 0.48]	0.40 [0.37, 0.43]	0.65 [0.63, 0.67]	0.54
Signature, refined charges	0.41 [0.36, 0.47]	0.71 [0.68, 0.74]	0.60 [0.57, 0.63]	0.55 [0.52, 0.58]	0.50 [0.47, 0.53]	0.69 [0.67, 0.70]	0.55
VADER, whole text	0.45 [0.40, 0.51]	0.72 [0.69, 0.75]	0.59 [0.55, 0.62]	0.45 [0.42, 0.48]	0.33 [0.29, 0.36]	0.55 [0.53, 0.57]	0.51
VADER, mean of sentence scores	0.46 [0.40, 0.51]	0.70 [0.67, 0.73]	0.57 [0.54, 0.61]	0.44 [0.41, 0.48]	0.31 [0.27, 0.34]	0.67 [0.65, 0.68]	0.50
Hedonometer	0.44 [0.39, 0.49]	0.62 [0.57, 0.66]	0.50 [0.46, 0.54]	0.34 [0.30, 0.38]	0.25 [0.21, 0.28]	0.60 [0.58, 0.62]	0.43
TextBlob	0.28 [0.20, 0.33]	0.51 [0.46, 0.56]	0.54 [0.49, 0.57]	0.42 [0.38, 0.45]	0.45 [0.42, 0.49]	0.64 [0.62, 0.65]	0.44
Bag-of-words ridge, other corpora	0.31 [0.25, 0.39]	0.63 [0.59, 0.67]	0.58 [0.54, 0.61]	0.56 [0.53, 0.59]	0.64 [0.62, 0.67]	0.68 [0.67, 0.70]	0.55
Bag-of-words ridge, same corpus	0.51 [0.46, 0.56]	0.58 [0.53, 0.62]	0.53 [0.49, 0.57]	0.67 [0.64, 0.69]	0.78 [0.76, 0.80]	0.85 [0.84, 0.85]	0.61
DistilBERT (SST-2)	0.61 [0.56, 0.63]	0.67 [0.63, 0.71]	0.62 [0.59, 0.65]	0.81 [0.80, 0.82]	0.62 [0.60, 0.65]	0.79 [0.78, 0.80]	0.67
Twitter-RoBERTa	0.75 [0.71, 0.78]	0.84 [0.81, 0.85]	0.83 [0.80, 0.84]	0.76 [0.74, 0.78]	0.30 [0.26, 0.33]	0.78 [0.77, 0.79]	0.70

Extended Data Table 3 Charge-weighted mean position of words, from -1 (first word) to $+1$ (last word), for texts of ten or more words, with 95% intervals; and the same for hedons and dolors separately.

Unit	Texts	All charge	Hedons	Dolors
EmoBank	5,975	+0.060 [+0.052, +0.068]	+0.051	+0.107
SemEval-2018 tweets	2,127	+0.105 [+0.096, +0.114]	+0.090	+0.156
Facebook posts	1,698	+0.021 [+0.012, +0.031]	+0.018	+0.036
SST sentences	9,358	+0.050 [+0.046, +0.055]	+0.042	+0.087
Film reviews	5,006	+0.023 [+0.022, +0.025]	+0.024	+0.022
Yelp reviews	157,452	+0.002 [+0.001, +0.002]	−0.001	+0.023
Film review paragraphs	22,514	+0.017 [+0.016, +0.019]	+0.011	+0.041
Novel chapters	2,583	+0.007 [+0.005, +0.009]	+0.005	+0.015

Extended Data Table 4 Coefficients of the shipped calibrated readout ($a = 4$, $\kappa = 0.5$), fitted on six corpora. The readout predicts a human valence rating in standard-deviation units.

Signature entry (per unit exposure)	Coefficient
intercept	−0.356
hedons	+5.850
dolors	−9.554
net charge $\times P_1$	+0.880
net charge $\times P_2$	+1.329
net charge $\times P_3$	+1.259
negated hedons	−15.196
negated dolors	+9.643
intensified net charge	+7.817
diminished net charge	−2.533
net charge after contrast	−0.470

Extended Data Table 5 Readings of the fitted coefficients, by fit. Dolor / hedon: how far a dolor moves the predicted rating relative to a hedon. Negated hedon, negated dolor: the multiple of its charge that a negated word counts for. Intensifier: the charge added, as a multiple of the modified word's. With b the coefficients of Extended Data Table 4, the four are $-b_{\text{dolors}}/b_{\text{hedons}}$, $1 + b_{\text{negated hedons}}/b_{\text{hedons}}$, $1 + b_{\text{negated dolors}}/b_{\text{dolors}}$ and $b_{\text{intensified}}$ divided by $(b_{\text{hedons}} - b_{\text{dolors}})/2$. The fit to film reviews alone is poorly determined.

Fit	Dolor / hedon	Negated hedon	Negated dolor	Intensifier
Shipped fit, six corpora ($a = 4$, $\kappa = 0.5$)	+1.63	−1.60	−0.01	+1.02
Other five corpora, EmoBank left out ($a = 2$, $\kappa = 0.2$)	+1.17	−1.19	−0.14	+1.08
Other five corpora, Tweets left out ($a = 4$, $\kappa = 0.5$)	+1.81	−1.95	−0.11	+1.06
Other five corpora, Facebook left out ($a = 4$, $\kappa = 0.5$)	+1.64	−1.79	−0.03	+0.99
Other five corpora, SST left out ($a = 4$, $\kappa = 0.5$)	+1.75	−1.75	+0.06	+1.09
Other five corpora, Film reviews left out ($a = 4$, $\kappa = 0.5$)	+1.63	−1.54	−0.02	+0.96
Other five corpora, Yelp left out ($a = 4$, $\kappa = 0.5$)	+1.57	−1.10	+0.10	+0.87
EmoBank alone ($a = 16$, $\kappa = 2$)	+2.22	−1.75	+0.19	+0.51
Tweets alone ($a = 2$, $\kappa = 0.5$)	+1.28	−0.51	+0.34	+0.93
Facebook alone ($a = 4$, $\kappa = 0.5$)	+1.64	−0.78	+0.07	+1.05
SST alone ($a = 1$, $\kappa = 0.5$)	+0.81	−0.69	−0.78	+0.66
Film reviews alone ($a = 32$, $\kappa = 0$)	−0.25	−1.23	+1.32	+7.89
Yelp alone ($a = 32$, $\kappa = 1$)	+4.48	−4.26	−0.39	+1.10

Extended Data Table 6 ChatGPT answers minus human answers in HC3, by source, in human standard deviations. For the four directional predictions (HL1 to HL4): no mark, supported in that source; †, refuted; ○, undecided (bootstrap over questions). The last two columns were not predicted, and the breakdown by source was added after review.

Source	Pairs	Net (HL1)	Dolors (HL2)	Clustering (HL3)	Ending lift (HL4)	Negated	Intensified
Reddit, explain like I'm five	14,009	+0.22	−0.10	+0.03†	+0.24	−0.24	−0.23
Finance	3,664	+0.13	−0.00○	−0.13	+0.08	−0.52	−0.31
Medicine	1,155	−0.74†	+0.28†	−0.45	−0.55†	−0.11	−0.07
Wikipedia, computer science	824	+0.14	−0.13	−0.10	+0.34	+0.03	+0.11
Open questions	439	+0.03○	−0.01○	+0.30○	+0.08○	+0.12	+0.10
All sources	20,091	+0.14	−0.06	−0.04	+0.13	−0.26	−0.21

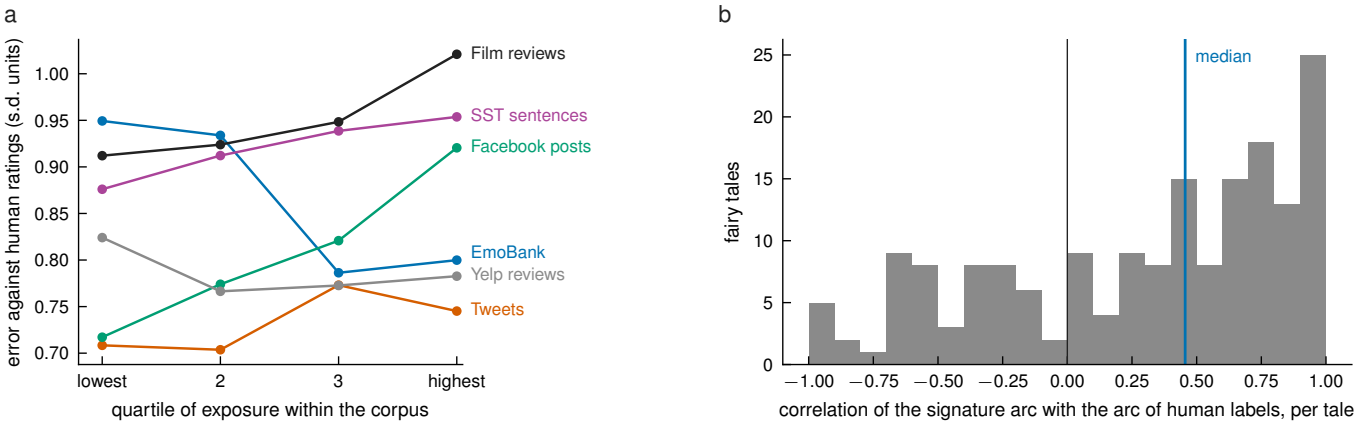
Extended Data Table 7 Charge of 20 words about groups of people and public figures in the human word norms and after refinement on rated text (refinement fitted on all six corpora; not released). The scale runs from −1 to +1.

Word	Norm	Refined	Word	Norm	Refined
women	+0.53	−0.08	*black*	−0.03	−0.07
men	+0.23	+0.09	*white*	+0.32	+0.14
girls	+0.48	+0.01	*asian*	+0.33	+0.01
boys	+0.29	+0.09	*mexican*	+0.10	+0.10
gay	−0.14	−0.39	*immigrants*	+0.06	+0.21
lesbian	+0.04	+0.03	*obama*	+0.23	−0.53
muslim	−0.09	−0.37	*trump*	+0.07	−0.14
islam	−0.08	−0.39	*old*	−0.26	−0.30
christian	+0.33	+0.25	*disabled*	−0.55	−0.63
jewish	+0.18	+0.26	*poor*	−0.61	−1.00

Extended Data Table 8 The dial at work on one essay, chosen as the essay at the median improvement in gap to its target (model: Sonnet; assignment: Seeking multiple opinions). First and last 45 words of the machine draft, of its revision after two rounds with the dial, and of its revision after two plain rounds; and five of the measures with the assigned target.

AI first draft	After two rounds with the dial	After two plain revisions
Opening. When I had to choose between two summer programs last year, my first instinct was to ask my mom and go with whatever she said. She thought the robotics camp was a safe bet because it looked good for college. But then I talked to … *Closing.* … it. But asking a few trusted people, especially people with different experiences, gives you more information, exposes hidden bias, and sharpens your own thinking. That is why I would always rather ask several people than rely on one, even when the first answer sounds convincing.	*Opening.* When I had to choose between two summer programs last year, my first idea was to ask my mom and do whatever she said. She thought the robotics camp was a safe choice because it would look great for college. Then I talked to my … *Closing.* … the end the choice belongs to the person who has to live with it. I feel glad I did not ask just one person. Kids who ask a few trusted people get a fair chance to choose, and that makes for a fun, wonderful summer.	*Opening.* Last summer I had to choose between two summer programs. At first I just wanted to ask my mom and do whatever she said. She told me the robotics camp was the safe choice because it would look good on my college applications. But then … *Closing.* … you ask a few people you trust, especially people who have had different experiences, you get more information and you can spot bias. That is why I would always ask a few people instead of just one, even if the first answer sounds really good.

Measure	Draft	Target	Dial	Plain
Hedons per 1,000 words	90.1	105.6	113.5	94.3
Dolors per 1,000 words	4.8	15.1	6.3	4.5
Charge negated (%)	4.7	7.3	7.6	5.7
Charge intensified (%)	0.8	2.0	0.0	3.7
Ending lift (per 1,000 words)	8.2	−19.7	12.7	−0.2



Extended Data Fig. 1 Error against exposure, and arcs of fairy tales. a, Root-mean-square error of the calibrated score against human ratings on each test portion, in standard-deviation units, by quartile of exposure within the corpus. **b**, Correlation between the cubic arc of a tale's merged signature and the cubic arc of its sentence labels from two human annotators, for 176 fairy tales.